CONFIDENT AND WRONG

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A hundred and twenty predictions made by people who knew better, and what the mistake was made of.

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Confident and Wrong

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END MATTER

BEFORE YOU BEGIN

How to Use This Book

This is a book of a hundred and twenty predictions that were wrong, made by people who were in an unusually good position to be right.

Each entry has four parts. The prediction, as it was made. Who made it, and when. What actually happened. And, in the last paragraph and most importantly, what the error was made of, which is the only part of any of this that is transferable.

It is not a book about idiots. That is the first thing to establish, because the genre is full of collections that are, and they are worthless. The physicist who said flight was impossible was the most eminent physicist in Britain and was reasoning correctly from the engines that existed. The company that turned down the telephone was the largest communications business in the world. The economist who announced a permanently high plateau eight days before the crash of 1929 was the leading monetary theorist of his generation, and he lost his own house. If your reaction to any entry is that the person was a fool, the entry has failed, and the note underneath it is where the actual finding is.

On hindsight. Everything here is unfair, and there is no way round it. You know the answer, which makes the reasoning look obvious. It was not obvious. There is good experimental evidence that once you know an outcome you cannot recover your genuine uncertainty about it. Read each prediction once before you read what happened, and try to notice whether you would have signed it.

On survivorship. These are the errors that left evidence. A collection of predictions that were wrong will always look as though experts are unreliable, because a prediction that came true is not interesting enough to record. The honest summary of the research is in the last entry of the book, and it is duller and more useful. Forecasters with strong theories do a little better than chance and are extremely confident. The ones who do best hold their views loosely and are never invited on television.

How it is arranged. Four parts. The Future, about things that could not be built and markets that would not exist, and its mirror, the promised futures that never arrived. The Settled Question, about science, medicine and engineering. Money. And People, about war, politics and the last chapter, which is about you.

On the quotations. Where a line is in quotation marks it is as recorded. Where it is in bold without quotation marks it is a summary of a documented position, because the wording is not reliably established. Where the attribution itself is doubtful, and several of the most famous lines in this genre are, the entry says *attributed to*. The note at the back explains which and why. The fakes are worth knowing about on their own account; the good ones are too neat, and the neatness is the tell.

Chapter twelve is the point. Everything before it is other people. The last chapter is about ordinary estimates made by competent professionals, and about the four things that are known to help. It is less entertaining than the eleven chapters in front of it. That is the correct order to read them in.

I

THE FUTURE

CHAPTER 1

Flight, and Other Impossibilities

Almost nobody in this chapter was a fool. That is the difficulty with the whole subject. It is worth facing on the first page and not discovering in the fifth. The people who said flight was impossible were, for the most part, the people best qualified to say anything about it at all.

There is a reason for that. It is not stupidity. Expertise is a detailed knowledge of obstacles. A man who has spent thirty years on steam engines knows, in his hands and not merely on paper, exactly how much a horsepower weighs. When he tells you that no engine can lift itself, he is not guessing. He is reporting, with precision, on every engine that exists.

What he cannot report on is the engine that does not exist yet. The obstacles are vivid, countable and present. The workaround is invisible until somebody finds it. So the informed estimate is not merely wrong. It is wrong in a particular direction, over and over, for a reason that never goes away.

The outsider does better in these pages, and it is worth being honest about why. He is not smarter. He is wrong far more often, about far more things. Nobody records it, because nobody asked him. Survivorship does the rest. This book is a collection of famous mistakes, and a book of the amateur's forecasts would be four times as thick and half as interesting.

Watch for one particular shape as you go. Time after time, the prediction is not *this cannot be done*. It is *this cannot be done by any means I can currently think of*, with the second half quietly dropped. That elision is the single most productive error in the history of forecasting, and it is still being made this week, probably by you. The men in this chapter had between them a Nobel prize, a presidency of the Royal Society, command of the largest research effort in American history and the senior uniform of the United States armed forces, and every one of them was overtaken inside a decade.

001

"Heavier-than-air flying machines are impossible."

Lord Kelvin, 1895

WHAT HAPPENED

Eight years. The Wright Flyer left the ground at Kitty Hawk. Within Kelvin's own lifetime, aeroplanes were being sold; within fifteen years of his death they were being manufactured in thousands for war. Kelvin was no crank. He was the most eminent physicist in Britain and president of the Royal Society. The temperature scale is named after him. He made the remark in the year he took that presidency.

WHAT THE ERROR WAS MADE OF

Reasoning from the engines that existed. Kelvin knew the power-to-weight ratio of steam, and by that arithmetic he was right: no steam plant of 1895 could lift itself and a man. What he treated as a law of nature was a fact about metallurgy and fuel. The internal combustion engine changed the constant, and the argument, which had been sound, did not survive it by a day.

002

"The demonstration that no possible combination of known substances, known forms of machinery and known forms of force, can be united in a practical machine by which man shall fly long distances through the air, seems to the writer as complete as it is possible for the demonstration of any physical fact to be."

Simon Newcomb, 1903

WHAT HAPPENED

Ten weeks, as it turned out. Newcomb published that in October 1903. The Wrights flew on the seventeenth of December, ten weeks later. He spent some of his remaining years arguing that what they had done did not count, since it could not carry a passenger. Then he argued that it never would.

WHAT THE ERROR WAS MADE OF

The escape clause he wrote himself and then forgot. *Known substances, known forms of machinery, known forms of force.* He stated the limits of his claim with perfect accuracy. Then he called it a demonstration of a physical fact. Nothing else in the sentence is wrong. The whole failure lives in a gap. What he established is that nobody yet knew how. What he thought he had established is that it could not be done. Say what you have shown.

003

"It might be assumed that the flying machine which will really fly might be evolved by the combined and continuous efforts of mathematicians and mechanicians in from one million to ten million years."

The New York Times, 9 October 1903

WHAT HAPPENED

Sixty-nine days. The editorial was prompted by Samuel Langley's aerodrome, which fell into the Potomac twice. From two failures by one man, the paper concluded that the thing itself was hopeless for geological ages.

WHAT THE ERROR WAS MADE OF

A number chosen for rhetoric and then believed. Nothing in the piece supports a million years over a thousand or ten; the figure is there to close the argument, not to carry it. Watch for this in your own arguments. A specific number is disarming, and the specificity of an invented one is exactly as persuasive as the specificity of a measured one.

004

"Man will not fly for fifty years."

Wilbur Wright, 1901

WHAT HAPPENED

Two years. He said it to his brother, in frustration, on the train home from a bad season of gliding at Kitty Hawk. He was flying two years later. He did not believe himself. He later recalled the remark with some amusement, as a man who had predicted his own failure and then gone home and got on with the work anyway.

WHAT THE ERROR WAS MADE OF

Mood, mistaken for judgement. It is the cheapest and commonest error in the book, and the only one here made by somebody who was actively solving the problem at the time. Notice what saved him. He did not act on it. A forecast made at the bottom of a bad week is a description of the week.

005

"Airplanes are interesting toys but of no military value."

attributed to Ferdinand Foch, 1911

WHAT HAPPENED

He commanded the war. Foch became the supreme commander of Allied forces in a war in which aircraft flew reconnaissance, artillery spotting, ground attack and strategic bombing, and he used them. Whether he said it is doubtful. It is one of the most-quoted lines in the genre and the sourcing is thin, so it is marked as attributed here and discussed in the note at the back.

WHAT THE ERROR WAS MADE OF

Judging a technology by its first version. In 1911 an aeroplane was a fabric kite that could carry one man and no cargo in fair weather, and the appraisal was accurate. The mistake, if he made it, was to assess the artefact instead of the rate at which it was improving. That rate is the most useful thing to look at. Almost nobody looks at it.

006

"Men might as well project a voyage to the Moon as attempt to employ steam navigation against the stormy North Atlantic Ocean."

Dionysius Lardner, 1835

WHAT HAPPENED

Three years. The Great Western crossed under steam in 1838. Lardner's argument was about coal: a ship large enough to carry fuel for the crossing would be too large to be driven by the engines of the day. Isambard Kingdom Brunel had already worked out what Lardner had not. A hull's capacity grows as the cube of its dimensions. Its drag grows as the square. Bigger ships have proportionally more room for coal, not less.

WHAT THE ERROR WAS MADE OF

A scaling law applied in the wrong direction. This is a subtler failure than the others here and it is the one that recurs most in engineering. Lardner had the right variables and got the relationship between them backwards, which is worse than ignorance, because it produces a confident number instead of a shrug.

007

Rockets cannot work in the vacuum of space, since they have nothing to push against.

The New York Times, 13 January 1920

WHAT HAPPENED

Forty-nine years, and then a correction. The editorial ridiculed Robert Goddard, then a professor at Clark University, for supposing a rocket could work outside the atmosphere. He seemed, it said, to lack the knowledge ladled out daily in high schools. Goddard was right and the newspaper was wrong about Newton's third law, which is high school physics. On 17 July 1969, as Apollo 11 travelled to the Moon, the paper printed a correction. Goddard had been dead for twenty-four years.

WHAT THE ERROR WAS MADE OF

Contempt used as a substitute for checking. The tone is the tell. An anonymous writer, secure in a general education, was certain he stood on the schoolboy's side of the argument, and never troubled to work out which side that was. Nobody is protected from this by being clever. The protection is the two minutes it takes to look. Two minutes.

008

“Space travel is utter bilge.”

Richard Woolley, 1956

WHAT HAPPENED

He lived to watch all of it. Woolley was the Astronomer Royal, and he said it to the press in his first week in the post. Sputnik went up eighteen months later, Gagarin five years after that, Apollo 11 thirteen years after the remark. Woolley, who lived until 1986, watched all of it. He argued afterwards, with some justice, that he had been answering a narrow question about the cost of a manned lunar programme. He was then quoted as though he had denied that rockets could leave the atmosphere.

WHAT THE ERROR WAS MADE OF

A short sentence eating a long argument. His considered position, which was that manned spaceflight would be enormously expensive and scientifically thin compared with instruments, has held up better than almost anything else in this chapter. But he said five words to a reporter, and the five words are what survived. If you are ever asked for a quotable line about something you actually understand, this is the entry to remember.

009

“I think we can leave that out of our thinking. I wish the American public would leave that out of their thinking.”

Vannevar Bush, 1945

WHAT HAPPENED

They were building it already. Bush was testifying to a Senate committee in December. A 3,000-mile rocket carrying an atomic warhead and landing inside a city, he said, was not going to be built for a very long time. The Soviet R-7 flew in 1957, twelve years later, and the American Atlas the year after. Bush was not a bystander: he had run the entire American wartime research effort, including the administrative machinery that produced the bomb he was talking about.

WHAT THE ERROR WAS MADE OF

Adding the difficulties together. Each piece of the problem was hard: the engine, the guidance, the re-entry vehicle, the warhead. Bush knew how hard. He had managed the people working on them. What he did was sum the obstacles when they were in fact being attacked in parallel by two states with unlimited budgets and an existential motive. Effort collapses timelines in a way that inventories of difficulty cannot show. Difficulties do not add.

010

"That is the biggest fool thing we have ever done. The bomb will never go off, and I speak as an expert in explosives."

William Leahy, 1945

WHAT HAPPENED

It went off. Leahy said it to President Truman. Trinity was detonated in July, Hiroshima in August. Leahy was Truman's chief of staff and the most senior military officer in the United States, and his objection was recorded by Truman himself. He remained, to his credit, a consistent critic of the weapon's use afterwards, on grounds that had nothing to do with whether it worked.

WHAT THE ERROR WAS MADE OF

Expertise in the neighbouring field. Leahy really was an expert in explosives, and everything he knew about chemical explosives was true and irrelevant. The clause *and I speak as an expert* is doing all the work, and it is the part that should have warned him. Authority crosses a boundary far more easily than the knowledge it was built on.

CHAPTER 2

Nobody Will Want One

The previous chapter was about whether a thing could be built. This one is about whether anybody would want it, and the failure rate is higher, because the question is worse.

A physicist denying flight is at least reasoning about something real. A businessman denying a market is reasoning about the future preferences of people who have never seen the object and cannot picture it. Asked, they will report that they are content with what they have. They are telling the truth when they say it. They are also, routinely, wrong about themselves, which is why market research on things that do not exist yet is close to worthless.

The second thing at work here is worse and more interesting. Almost every man in this chapter was an incumbent. He had built something, he understood it in detail, and his understanding was organised around the thing he had built. That is not a bias in the ordinary sense. It is a structure. He can see the new object perfectly well. What he cannot see is the world in which it matters, because his idea of the world is the one his own product fits into.

Notice how few of these predictions are stupid at the moment they were made. The iPhone in 2007 was expensive, slow on the network of the day, had no keyboard, and sold to nobody outside a rich niche. Everything Steve Ballmer said about it was true. It was true for about eighteen months.

That is the shape to watch for. A market denial is nearly always accurate as a description and wrong as a forecast. It describes the object and not the slope it is on. Every one of these men had a spreadsheet, a market, a sales force and a decade of correct forecasts behind him, and what none of them had was a way of pricing a product whose whole significance lay in a market that did not exist yet.

011

“There is no reason for any individual to have a computer in his home.”

Ken Olsen, 1977

WHAT HAPPENED

Fifteen years. Olsen was the founder of Digital Equipment Corporation, then the second-largest computer company in the world. Within fifteen years the personal computer had destroyed the minicomputer business his company was built on, and in 1992 the board removed him. He spent decades afterwards explaining, with real irritation and some justice, that he had meant a computer that ran the house and turned the lights on and off. He had not meant a machine you would work on.

WHAT THE ERROR WAS MADE OF

A defensible sentence detached from the sentence before it. His defence is probably true, and it did not save him, and it never does: a line that can be quoted alone will be. The deeper problem is the question he was answering. *What would a home computer be for* got settled out of his own idea of what a computer was for, which was serious institutional work. The thing that killed his company was not a better minicomputer. It was a toy that got good.

012

"While theoretically and technically television may be feasible, commercially and financially I consider it an impossibility, a development of which we need waste little time dreaming."

Lee de Forest, 1926

WHAT HAPPENED

Ten years. De Forest had invented the triode valve, the component that made both radio and television electronics possible, and is one of the genuine founding figures of the field. Regular television broadcasting began in Britain in 1936, ten years later. By the time he died in 1961 it was the dominant medium on earth. He revised his view long before that and said so.

WHAT THE ERROR WAS MADE OF

A conflation of two different words. He separated *technically feasible* from *commercially possible*, which is exactly the right distinction and better than most people manage. Then he treated the commercial question as if it had a fixed answer, when it is a question about costs that fall every year. Almost nothing is commercially impossible. Things are commercially impossible at a price, and the price is a moving object. Ask at what price.

013

"Television won't be able to hold on to any market it captures after the first six months. People will soon get tired of staring at a plywood box every night."

Darryl Zanuck, 1946

WHAT HAPPENED

He had it exactly backwards. Zanuck ran Twentieth Century Fox. American cinema admissions peaked in 1946, the year he said it, and then fell by more than half over the following decade as television arrived in the home. His own industry spent the 1950s inventing widescreen formats specifically to give audiences something the plywood box could not.

WHAT THE ERROR WAS MADE OF

Contempt for the medium instead of attention to the friction. Television in 1946 was worse than cinema in every way he could name, and he named them accurately. What he did not weigh was the only variable that mattered, which is that it was in the house. Convenience beats quality far more often than people in the quality business can bring themselves to believe.

014

"Remote shopping, while entirely feasible, will flop, because women like to get out of the house, like to handle merchandise, like to be able to change their minds."

Time, 1966

WHAT HAPPENED

It did not flop. It took thirty years to begin and another twenty to become ordinary, so the prediction had a long and comfortable retirement before it failed. Online retail passed a fifth of all American retail sales within sixty years of the piece.

WHAT THE ERROR WAS MADE OF

A real observation turned into a permanent law of human nature. People genuinely did like handling merchandise, and many still do. The error is in the word *because*. A preference is a matter of degree, traded off against other things, and the sentence promotes it to a fixed fact about a whole category of person. Watch for a forecast that rests on what people are like. People are like that until something is cheap and easy enough that they are not.

015

"The real future of the laptop computer will remain in the specialised niche markets. Because no matter how inexpensive the machines become, and no matter how sophisticated their software, I still can't imagine the average user taking one along when going fishing."

Erik Sandberg-Diment, 1985

WHAT HAPPENED

He was wrong twice over. Laptops passed desktops in sales in 2008, and the pocket version, which he was not even contemplating, went fishing with something like four billion people.

WHAT THE ERROR WAS MADE OF

The words *I still can't imagine*, offered as evidence. He is transparent about his method, which is more than most people in this book manage, and his method is to consult his own imagination and report the result. The failure of imagination is not a fact about the world. It is a fact about the imagination, and the honest version of that sentence ends with a question rather than a conclusion. Imagination is not evidence.

016

“Visionaries see a future of telecommuting workers, interactive libraries and multimedia classrooms. They speak of electronic town meetings and virtual communities. Commerce and business will shift from offices and malls to networks and modems. Baloney.”

Clifford Stoll, 1995

WHAT HAPPENED

All of it happened. All of it happened, most of it within fifteen years, and the pandemic of 2020 delivered the telecommuting item in a fortnight. Stoll has been unusually good about this since, reprinting the piece himself and pointing out that he was wrong on almost every count.

WHAT THE ERROR WAS MADE OF

Correct observations about the present, dated by an argument about the future. His specific complaints were sound. The network of 1995 was slow, the search was terrible, and there was no way to pay for anything. Each was a fixable engineering problem. He read them as evidence about what the medium was, and not as a list of what had not been done yet. The list was, in effect, a roadmap, and he had written it himself.

017

“I predict the Internet will soon go spectacularly supernova and in 1996 catastrophically collapse.”

Robert Metcalfe, 1995

WHAT HAPPENED

It did not. Metcalfe invented Ethernet and is not a man who misunderstands networks. He had promised to eat his words if he was wrong. In 1997 he put a printed copy of the column in a blender at an industry conference and drank it in front of the audience.

WHAT THE ERROR WAS MADE OF

Extrapolating a failure mode past the point where anyone would notice and fix it. He was right about the congestion, right about the brownouts, right that the growth curve was outrunning the infrastructure. What he modelled was a system nobody was maintaining. Almost every apocalyptic forecast about an operating system, a network or an institution shares this shape: it assumes the operators will watch it happen. Somebody is watching the graphs.

018

"The growth of the Internet will slow drastically. By 2005 or so, it will become clear that the Internet's impact on the economy has been no greater than the fax machine's."

Paul Krugman, 1998

WHAT HAPPENED

The network held. It did not slow, and the internet's economic footprint by 2005 was larger than any single technology of the twentieth century. Krugman won the Nobel prize in economics ten years after saying it. A wrong forecast tells you very little about a mind.

WHAT THE ERROR WAS MADE OF

The Metcalfe's law argument, borrowed and misused. His reasoning was that most people have nothing to say to one another, so the value of connecting everybody had been overstated. That is a real and underrated point about network value, and it is the correct explanation for why several later social products failed. He aimed it at the network itself and not at the applications running on it. A general argument fired at the wrong level of a stack sounds rigorous and lands nowhere.

019

“What would I do? I’d shut it down and give the money back to the shareholders.”

Michael Dell, 1997

WHAT HAPPENED

Ten years. Apple was ninety days from insolvency when he said it. Ten years later it passed Dell in market value, and it is now worth something on the order of fifty times as much. Steve Jobs sent the remark to every employee and put it on a screen behind him at a company meeting, which is one way of using a competitor’s opinion.

WHAT THE ERROR WAS MADE OF

Valuing a company by its balance sheet rather than by what was about to be put into it. Dell’s judgement of Apple in 1997 was correct in every particular that could be counted. He had no way to count the thing that mattered: a returning founder and four unreleased products. This is the standard limitation of financial analysis and not a scandal. The scandal is only in the confidence. Count what can be counted, and say so.

020

“There’s no chance that the iPhone is going to get any significant market share. No chance.”

Steve Ballmer, 2007

WHAT HAPPENED

It got the share. By 2012 the iPhone alone was generating more revenue than the whole of Microsoft. Windows Mobile, the product Ballmer was defending, was discontinued. He has been generous about it since. At the time he was giving the correct summary of the business as it existed: no keyboard, no enterprise support, one carrier, six hundred dollars.

WHAT THE ERROR WAS MADE OF

Measuring against the wrong customer. Ballmer's market was corporate IT departments buying handsets in bulk, and for that customer everything he said was true. The iPhone did not compete for that customer. It went to the individual, who took it to work and made the IT department deal with it. What had been a weakness in the first market was the whole product in the second.

CHAPTER 3

The Ones That Never Arrived

If the first two chapters were the whole story, the moral would be simple and false: bet on the new thing, always, because the doubters are always wrong.

They are not. This chapter is the other half of the ledger, and it is longer than anyone remembers, because a thing that did not happen leaves nothing behind to remind you of it. The failed prophecy of doom has no wreckage. The promised technology that never shipped has no museum. We remember Kelvin on flight because there are aeroplanes to remind us; nobody has a nuclear vacuum cleaner in the cupboard to jog the memory about Alex Lewyt.

So the two errors are not equally visible, and the imbalance in the record is not an imbalance in the world. Optimism and pessimism about the future fail at similar rates. Optimism just fails more quietly.

There is a family resemblance across the entries here. Nearly all of them take a problem that has been solved in the laboratory and assume the remaining work is a formality. It almost never is. The demonstration is the easy part. The hard part is cost, reliability, regulation, habit, and the tedious business of getting several thousand ordinary things to work at once. None of it is interesting enough to write about and all of it takes decades.

The second resemblance is a matter of arithmetic. A forecaster who says twenty years is usually describing a feeling, and the feeling is *this is clearly coming and I want to be alive for it*. Twenty years is the horizon at which a prediction is bold enough to be worth making and distant enough that nobody will check. Machine intelligence has been twenty years away for seventy years, and fusion has been thirty away for eighty. The regularity is useful enough to serve as an instrument: whenever somebody attaches twenty years to a transformation, ask what has to happen in years three, seven and fourteen, and watch how quickly the answer becomes a wave of the hand.

021

Population grows geometrically while food supply grows arithmetically, so want and famine are permanent and unavoidable.

Thomas Malthus, 1798

WHAT HAPPENED

It did not happen. World population has gone up roughly eightfold since he wrote, and the proportion of people who are hungry has fallen to the lowest level in recorded history. Famine still occurs, and where it occurs it is now nearly always a political event rather than an agricultural one. Malthus was not answered by an argument. He was answered by nitrogen fertiliser, mechanisation, and plant breeding.

WHAT THE ERROR WAS MADE OF

Two trend lines drawn on the assumption that neither responds to the other. It is the founding error of the genre and every entry in this chapter is a variation on it. Malthus had no reason to expect the Haber process. He had every reason to expect that the pressure of hunger would drive people to work furiously on the problem of yield, which is exactly what happened. A model of the future that leaves out the reaction to the model is describing a world without people in it.

022

"Three-hour shifts or a fifteen-hour week may put off the problem for a great while. For three hours a day is quite enough to satisfy the old Adam in most of us."

John Maynard Keynes, 1930

WHAT HAPPENED

The leisure never arrived. He predicted that by roughly 2030 his grandchildren's generation would work about fifteen hours a week, because productivity would have risen enough to make more unnecessary. The productivity arrived, more or less on the schedule he set out. The leisure did not. Working hours fell for about forty years and then stopped falling, and in some professions reversed.

WHAT THE ERROR WAS MADE OF

Treating wants as a fixed quantity to be satisfied. Keynes distinguished absolute needs from relative ones and then assumed the relative ones would run out of steam, which is the one part of the essay he does not argue for. Wants turned out to be manufactured as fast as the means of meeting them. A good deal of the productivity went into goods that did not exist in 1930, and not into afternoons off.

023

"Mark my word: a combination airplane and motorcar is coming. You may smile, but it will come."

Henry Ford, 1940

WHAT HAPPENED

It never came. Every decade since has produced a prototype, a magazine cover and a company that went under. The obstacles are not aeronautical: the machine can be built, and has been, many times. What kills it is simpler. A good car is a bad aeroplane and a good aeroplane is a bad car, and a country of amateur pilots in three dimensions over a city is a proposition no regulator will accept.

WHAT THE ERROR WAS MADE OF

Adding two solved problems together and assuming the sum is solved. Ford had done exactly this with the car itself, brilliantly, and it worked, which is presumably why he expected it to work again. Combination is not free. Almost every constraint that binds one half of a hybrid contradicts a constraint on the other. The result is worse at both jobs than the two separate machines it replaces.

024

"It is not too much to expect that our children will enjoy in their homes electrical energy too cheap to meter."

Lewis Strauss, 1954

WHAT HAPPENED

It is metered, and it is not cheap. The reactor fuel is nearly free, which is the part Strauss, then chairman of the Atomic Energy Commission, was thinking about. The plant that contains it is not, and the cost of building one has risen in real terms for sixty years, which is close to unique among manufactured things.

WHAT THE ERROR WAS MADE OF

Costing the fuel instead of the system. It is a specific and repeated failure and it has a shape you can learn to see: the expert prices the part he finds interesting and treats the rest as detail. Concrete, steel, litigation, insurance, decommissioning and the institutional weight of proving to a frightened public that a thing is safe turned out to be the whole cost of nuclear power. None of it was in his sentence.

025

“Nuclear-powered vacuum cleaners will probably be a reality within 10 years.”

Alex Lewyt, 1955

WHAT HAPPENED

No. Lewyt made vacuum cleaners for a living and was quoted in the New York Times, and the remark is now used, unfairly, as a punchline about the credulousness of the 1950s. It was a thoroughly mainstream opinion. The atom was new, it was small, and for about a decade the assumption that everything would be run by it was ordinary rather than eccentric.

WHAT THE ERROR WAS MADE OF

Applying a general technology to every particular without asking what the particular needs. A vacuum cleaner’s problem is not energy density: it plugs into a wall, and the wall has as much power as anybody could want. This is the standard failure around a genuinely important new technology, and it recurs with every one of them. The question worth asking is never *where could this be used* but *where is the existing solution actually painful*.

026

“Machines will be capable, within twenty years, of doing any work a man can do.”

Herbert Simon, 1965

WHAT HAPPENED

Sixty years on, they cannot, though the boundary has moved a very long way and continues to move. Simon was not a hobbyist. He won the Nobel prize in economics and the Turing award, and he was one of the founders of artificial intelligence as a field. He said, in the same period, that a computer would be chess champion of the world within ten years. That took thirty-two.

WHAT THE ERROR WAS MADE OF

Mistaking the hardest-looking part for the hard part. The early results were startling: within a decade machines could prove theorems and play a decent game, which are things clever humans find difficult. The reasonable inference was that the easy things would follow at once. What followed instead was the discovery that walking across a room, understanding a sentence and recognising a face are enormously harder than symbolic logic. The tasks we find effortless are the ones evolution spent longest on and the ones we can least explain.

027

"In the 1970s hundreds of millions of people will starve to death in spite of any crash programs embarked upon now."

Paul Ehrlich, 1968

WHAT HAPPENED

They did not. They did not. The 1970s saw famines, and they were regional and political. Over the same decade the Green Revolution roughly doubled cereal yields in India and Mexico. Norman Borlaug, who bred the wheat, is credited with a number of lives saved that runs into the hundreds of millions. Ehrlich later lost a public wager with the economist Julian Simon over whether the price of five metals would rise across the 1980s. All five fell.

WHAT THE ERROR WAS MADE OF

Malthus again, with better graphs. The book's method is to take the current yield and the current birth rate and run them forward. Its most confident passages are about the impossibility of the exact thing that was happening in Mexican wheat fields while it was being written. The specificity is what makes it instructive: a forecast phrased as *in spite of any crash programs embarked upon now* has explicitly ruled out the mechanism that then rescued everyone.

028

"In from three to eight years we will have a machine with the general intelligence of an average human being."

Marvin Minsky, 1970

WHAT HAPPENED

The field ran out of money instead. The funding was cut in Britain after the Lighthill report of 1973 and in America not long after. The people who lived through what followed call it the AI winter. Minsky was the leading figure in the field and co-founded the laboratory at MIT.

WHAT THE ERROR WAS MADE OF

A promise made in the currency of grant applications and then quoted as a forecast. There is a systematic distortion around any field that lives on funding, and it does not require anybody to lie. The optimistic version gets the money, so the optimistic version gets said. After a while it is what everybody in the room believes, because it is what everybody in the room has heard. Read any timeline from inside a field that is asking for money with that in mind, including the ones being published now.

029

The office of the future will do away with paper. Documents will be called up on a screen, and executives will keep almost nothing on a desk.

George Pake, 1975

WHAT HAPPENED

Paper consumption went up. American office paper consumption then rose for a quarter of a century, and peaked around the turn of the century, at roughly double the level of the forecast. Pake ran Xerox PARC, where the technology that eventually did reduce paper was being invented at that moment, so he was not wrong about the tools. He was wrong about the sequence.

WHAT THE ERROR WAS MADE OF

Forgetting that a new medium first makes the old one cheaper to use. Word processors and laser printers, both PARC products, made it trivial to produce a document, so people produced far more of them and printed most. The paperless office arrived eventually, thirty years late, and it arrived when screens got good and habits changed, not when the software did. A technology's first effect is almost always to amplify the thing it will later replace.

030

It will be to the car what the car was to the horse and buggy, and it may be more important than the internet.

John Doerr, 2001

WHAT HAPPENED

Around 140,000 were sold, over eighteen years, mostly to police forces, warehouses and tourists in single file. The company was bought by a Chinese scooter manufacturer and the original product was discontinued in 2020. The machine worked perfectly. That was never the question.

WHAT THE ERROR WAS MADE OF

Excitement about the engineering, in the absence of a person with a problem. The self-balancing mechanism was a genuine and beautiful piece of work, and the investors who saw it demonstrated were, by their own accounts, dazzled by it. Nobody in those rooms was a commuter deciding between this and a bicycle, at a price of five thousand dollars, on a pavement where it was illegal. The demonstration answers the question *does it work*, and the question that decides the outcome is *who is worse off without it*.

II

THE SETTLED QUESTION

CHAPTER 4

The Science That Was Closed

Science is the only human activity with a formal procedure for admitting error, which is why its famous mistakes are so well documented and so often misread. A discipline that keeps minutes will always look worse than one that does not.

The failures collected here are not failures of method. They are failures of timing and temperament. A correct procedure was applied by people who had decided in advance what the answer would be, or who could not see that a question was open because everybody they respected had stopped asking it.

Two shapes recur. The first is the declaration that a field is essentially finished, which appears at the end of every long run of success and is always followed by an upheaval. It is not a coincidence. A period of consolidation is exactly when the remaining anomalies look like rounding errors, and the anomalies are where the next century is.

The second is rejection by a discipline of something arriving from outside it. Wegener was a meteorologist telling geologists about continents. Marshall was an unknown registrar telling gastroenterologists about bacteria. In both cases the objection was framed as methodological rigour and functioned as territorial defence, and in both cases the outsider was right and had to wait decades.

The uncomfortable part is that the same instinct, applied to the far larger number of outsiders who are wrong, is correct and necessary. A discipline that admitted every amateur with a theory would drown in a week. There is no clean rule here. What there is, at best, is a habit of noticing when you are dismissing an argument because of who made it, and separating that from whether it is true. What makes these entries worth reading is not that clever people were wrong, which is unremarkable, but that in nearly every case the evidence which undid them was already on the table, published and reproducible and ignored, sometimes for forty years and once for very nearly a century.

031

Stones do not fall from the sky, because there are no stones in the sky.

The French Academy of Sciences, 1772

WHAT HAPPENED

Thirty-one years. The Academy, with Lavoisier among the signatories, investigated reported meteorite falls and concluded that the stones were terrestrial, probably struck by lightning. Museums across Europe threw away their meteorite collections to avoid embarrassment. In 1794 Ernst Chladni argued the stones came from space and was mocked. In 1803 some three thousand of them fell on the town of L'Aigle in Normandy in front of witnesses. The Academy sent Jean-Baptiste Biot to investigate and the matter was settled in a season.

WHAT THE ERROR WAS MADE OF

A prior belief strong enough to discard the evidence rather than the belief. Peasants had been reporting falling stones for centuries and were disbelieved on the grounds that peasants report all sorts of things. What is instructive is the direction of the reasoning: the Academy did not weigh the testimony and find it wanting. It knew there was nothing up there to fall, so the testimony had to be wrong, and the work of the investigation was to find out how.

032

We shall never be able by any means to study the chemical composition of the stars, or their mineralogical structure.

Auguste Comte, 1835

WHAT HAPPENED

Twenty-four years. In 1859 Kirchhoff and Bunsen showed that the dark lines in the solar spectrum identify the elements present in the sun. By the end of the century the composition of stars was a routine measurement. Helium was discovered in the sun before it was found on earth.

WHAT THE ERROR WAS MADE OF

Confusing the impossibility of going there with the impossibility of knowing. Comte's argument was that we can never visit a star and never bring a sample home, both of which are still true. He treated physical access as the only route to knowledge about a distant object. The whole subsequent history of astronomy is the discovery that light carries the information without anybody needing to travel. The lesson generalises: an impossibility argument usually rules out one method and quietly claims to have ruled out the goal.

033

Physics is essentially complete. There may be a speck of dust here and a bubble there to examine, but the system as a whole stands secure.

Philipp von Jolly, 1874

WHAT HAPPENED

Twenty-six years. Jolly said it to the young Max Planck, who was asking whether to study physics, and advised him to do something else. Planck went into it anyway. In 1900, to solve a problem about the radiation from a hot body that his teacher would have called a speck of dust, he introduced the quantum. Planck told the story himself decades later, without malice.

WHAT THE ERROR WAS MADE OF

Mistaking the size of the remaining anomalies for the size of their consequences. The unexplained residue in 1874 was small in every measurable sense: a discrepancy here, a stubborn curve there. That is what an unexplained residue looks like just before a revolution. A mature theory has already absorbed everything it can, and what is left over is by definition what it cannot absorb. Small anomalies in a strong theory are not a sign that the theory is nearly done. They are the only place a new one can come from.

034

“The more important fundamental laws and facts of physical science have all been discovered, and these are now so firmly established that the possibility of their ever being supplanted in consequence of new discoveries is exceedingly remote. Our future discoveries must be looked for in the sixth place of decimals.”

Albert Michelson, 1894

WHAT HAPPENED

Not the sixth decimal place. Radioactivity was found two years later, the quantum in 1900, special relativity in 1905. Michelson himself had already performed, with Edward Morley in 1887, the experiment whose null result relativity would explain. He won the Nobel prize in 1907 for the precision of his optical measurements. He never accepted relativity.

WHAT THE ERROR WAS MADE OF

A confusion between precision and completeness. Michelson was the greatest measurer of his generation, and from inside that discipline the frontier genuinely does look like the sixth decimal place, because that is where his work lived. His own most famous experiment was a sixth-decimal-place measurement that broke the theory it was designed to confirm. Being the person who produces the crucial anomaly is no protection at all against failing to see it.

035

The straight lines on Mars are canals, dug by an intelligent race to carry water from the poles across a drying planet.

Percival Lowell, 1895

WHAT HAPPENED

There are no canals. Lowell built an observatory in Arizona specifically to study them, published three books, and convinced a substantial part of the public. Better telescopes showed nothing. Mariner 4 flew past in 1965 and photographed craters. There are no canals. The lines were an artefact of the eye assembling faint, blurred detail into continuous features, a tendency later confirmed by experiments in which observers drew canals on featureless discs.

WHAT THE ERROR WAS MADE OF

Observation contaminated by the hypothesis it was testing. Lowell was not lying and he was not careless. He was an experienced observer looking at the edge of what his instrument could resolve, which is the regime in which the brain supplies what the data lacks. Other astronomers with better sites saw nothing, and he explained that away by the quality of their seeing. Any observation that requires exceptional conditions to reproduce, and gets stronger the more you expect it, should be treated as evidence about the observer.

036

A new kind of radiation, N-rays, is emitted by most substances and can be detected as a change in the brightness of a spark.

René Blondlot, 1903

WHAT HAPPENED

One year. Blondlot was a respected French physicist and the discovery followed close on X-rays, which had been real and astonishing. Within two years around a hundred papers were published on N-rays, mostly by French laboratories. In 1904 the American physicist Robert Wood visited Blondlot's laboratory, and in the dark removed the aluminium prism essential to the apparatus. Blondlot continued to report the readings. Wood published what he had done, and the field evaporated.

WHAT THE ERROR WAS MADE OF

A measurement whose only instrument was human judgement of a faint change. Every safeguard that matters was absent: no blinding, no control, no way for the observer not to know what he was supposed to see. The hundred confirming papers are the important detail. Replication by people who share your expectation and your method is not evidence. A result that appears only in the laboratories that believe in it has told you where the effect lives.

037

"If we are to believe Wegener's hypothesis we must forget everything which has been learned in the last seventy years and start all over again."

Rollin Chamberlin, 1926

WHAT HAPPENED

Forty years. Alfred Wegener proposed continental drift in 1912; the American geological establishment rejected it at a symposium in 1926, of which that sentence is the summary. Wegener died on the Greenland ice in 1930 with the question closed against him. Seafloor spreading was demonstrated in the early 1960s, plate tectonics was established by about 1968, and geology did in fact have to start again.

WHAT THE ERROR WAS MADE OF

A true objection allowed to settle a question it could not settle. Wegener had no mechanism: he could not say what moved a continent, and his own suggestions were wrong. That is a serious weakness and the geologists were right to press it. What they were not entitled to do was convert *we do not know how* into *it did not happen*. The coastlines fitted. The fossils matched, and so did the rock. Missing mechanism is a reason to keep looking. It is not a refutation, and the cost of treating it as one was forty years.

038

“Anyone who expects a source of power from the transformation of these atoms is talking moonshine.”

Ernest Rutherford, 1933

WHAT HAPPENED

One night in Bloomsbury. Leo Szilard read the report of the speech in a newspaper the following morning, was irritated by it, and while waiting at a traffic light in Bloomsbury conceived the nuclear chain reaction. He filed a patent the next year. The first self-sustaining reaction ran in 1942, and the first bomb was tested in 1945, twelve years after the remark. Rutherford had split the atom himself and knew more about it than anyone alive.

WHAT THE ERROR WAS MADE OF

Reasoning from the efficiency of the method he was using. Rutherford’s technique was to fire particles at a target and hope for a hit, and by that route the energy you put in vastly exceeds the energy you get out, for ever. His arithmetic was right. What he had not conceived of was a reaction that supplies its own particles, and once you have that idea the accounting inverts completely. A rate argument is only as good as the mechanism it assumes, and a new mechanism does not respect it.

039

The universe has always existed, and matter is created continuously as it expands; the notion of everything beginning in a single explosion is not serious cosmology.

Fred Hoyle, 1949

WHAT HAPPENED

Sixteen years. Hoyle coined the phrase *big bang* on the radio, apparently as a description rather than an insult, and it stuck to the theory he was attacking. The cosmic microwave background was found in 1965 and is very difficult to explain in a steady-state universe. Hoyle never accepted it, and argued against it for the remaining thirty-five years of his life. He was also, in the same period, the principal author of the theory of how stars manufacture the elements, which is one of the great achievements of twentieth-century science and is correct.

WHAT THE ERROR WAS MADE OF

A metaphysical preference defended with technical skill. Hoyle disliked a beginning, partly on the grounds that it smuggled in a creation, and he had the ability to construct a serious alternative and to keep repairing it. The tell is that the theory kept needing repairs while the other one kept receiving unforced confirmations. A position that survives only by amendment, held by somebody clever enough to keep amending it, can last a lifetime.

040

We have achieved sustained nuclear fusion at room temperature in a benchtop electrochemical cell.

Martin Fleischmann and Stanley Pons, 1989

WHAT HAPPENED

Nobody could reproduce it. They announced it at a press conference, before publication, and the world's laboratories spent several months and a great deal of money failing to reproduce it. The excess heat was not confirmed and the neutron signature, which fusion would require, was not there. Both were serious electrochemists, and neither recovered professionally.

WHAT THE ERROR WAS MADE OF

An announcement made ahead of the checking, in a race. They believed a rival group was about to publish, and the press conference was a claim to priority. Everything that would normally have caught the error, peer review, blinded replication, the slow argument with sceptical colleagues, was scheduled to happen after the world had been told. The order matters more than the care: it is not that they were sloppier than other scientists, it is that they inverted the sequence in which sloppiness gets found.

CHAPTER 5

Doctors, Certain

Medicine has a problem the other sciences do not. The patient in front of you cannot wait for the question to be settled. You have to act on the best available belief, and acting on it commits you to it in a way that reading about it does not.

A physicist who backs the wrong theory has wasted some years. A physician who backs the wrong theory has bled four hundred people. That is not an argument for timidity, because refusing to act is also a choice with a body count, and the doctors in this chapter were mostly not cowards. It is an argument for noticing how much harder it becomes to change your mind once you have done the thing.

The second difficulty is that almost everybody gets better. Most conditions resolve, most of the time, whatever you do, and a treatment applied to a self-limiting illness will appear to work. Every generation of physicians has watched patients recover under a therapy we now know to be useless or actively harmful, and has taken the recovery as evidence. The randomised controlled trial exists precisely because clinical experience, which feels like the most reliable thing in the world, is not reliable at all.

Three of the entries below are not really errors of reasoning. They are cases where a company knew, or had good reason to suspect, and kept selling. They are here because from the outside the two look identical. The machinery that lets an honest person stay wrong is the machinery a dishonest one uses on purpose.

A note before you start. It is very easy to read this chapter and conclude that medicine cannot be trusted. The correct conclusion is the opposite and duller: this is what a field looks like when it audits itself. Every entry here was found out by doctors. It is worth holding on to as you read, because the bleeding, the lobotomy, the radium tonic and the drug that passed every test anybody had thought to run were not the work of an outside quackery, but of the profession's most decorated men, working in its best hospitals on its own authority.

041

Yellow fever is best treated by copious bleeding, repeated as often as the symptoms return.

Benjamin Rush, 1793

WHAT HAPPENED

His patients died. Rush signed the Declaration of Independence and was the most famous physician in America. He treated the Philadelphia epidemic by removing enormous quantities of blood, sometimes several pints over a few days, along with violent purging. Mortality among his patients was appalling. He believed to the end of his life that he had saved the city, and he sued a journalist who said otherwise, and won.

WHAT THE ERROR WAS MADE OF

A theory that could absorb any outcome. Rush held that fever was a disorder of excessive arterial action, so recovery proved the bleeding had worked and death proved it had come too late or been too timid. There is no result the patient could produce that would count against it. The single most useful question you can put to any strongly held position is what observation would change it, and the answer here, for twenty years, was none. Rush never asked it.

042

“Doctors are gentlemen, and gentlemen’s hands are clean.”

attributed to Charles Meigs, 1854

WHAT HAPPENED

It took eighteen years and a death. In 1847 Ignaz Semmelweis, working in a Vienna maternity ward, noticed that the wing staffed by doctors killed mothers at several times the rate of the wing staffed by midwives. The doctors came to the deliveries from the dissecting room. He instituted handwashing in chlorinated lime and the death rate collapsed. The profession rejected him, in Vienna and abroad. He was dismissed, deteriorated, and died in an asylum in 1865 at forty-seven, of an infection. Lister’s antiseptic work vindicated him within a decade.

WHAT THE ERROR WAS MADE OF

An accusation heard instead of a hypothesis. Semmelweis was telling obstetricians that they were killing their own patients with their hands, which was true and unbearable. He had no germ theory to explain it with. What he offered was a rebuke without a mechanism. His own manner made it worse; he wrote to his critics calling them murderers. Being right is not a strategy, and a finding that requires its audience to accept guilt will be tested more harshly than one that does not.

043

Cocaine is a harmless stimulant, useful against depression, indigestion, morphine addiction and alcoholism, with no habituation of its own.

Sigmund Freud, 1884

WHAT HAPPENED

It ruined a friend of his. Freud published *Über Coca*, took the drug himself, and gave it to his close friend Ernst von Fleischl-Marxow to break a morphine habit. Fleischl-Marxow became one of the first documented cocaine addicts in

Europe and died at forty-five. Freud quietly withdrew from the subject and the episode damaged his reputation for a decade.

WHAT THE ERROR WAS MADE OF

A sample of one, and the one was himself. He felt better on it, which is what the drug does, and he reported the feeling as a pharmacological finding. The trap is particular. A substance that alters your judgement is being evaluated by your judgement, and the better it is at what it does, the more favourable the report. Personal experience is the weakest evidence available in precisely the cases where it feels the strongest. He was the experiment.

044

Radithor is harmless in every respect. A dose of certified radioactive water taken daily restores energy and cures more than one hundred and fifty conditions.

William Bailey, 1928

WHAT HAPPENED

It killed its best customer. Bailey, who was not a doctor and had no degree, sold radium dissolved in water by mail order. Eben Byers, a wealthy Pittsburgh industrialist, drank around fourteen hundred bottles on his physician's advice. His jaw disintegrated and had to be removed, holes opened in his skull, and he died in 1932. He was buried in a lead-lined coffin. The Federal Trade Commission acted, and the case is one of the reasons the American food and drug regulator has the powers it has.

WHAT THE ERROR WAS MADE OF

A new invisible force adopted as a general tonic before anybody had asked what it does to tissue. Radium was known to destroy cells, which was why it was being used against tumours. The leap to swallowing it daily meant treating an established destructive effect as a stimulating one at low doses, with no evidence and no reason. Whenever something is being sold as good for a hundred and fifty unrelated conditions, the number is the diagnosis.

045

Cutting the connections to the frontal lobes relieves severe mental illness and should be used widely.

Walter Freeman, 1942

WHAT HAPPENED

It was abandoned within a decade. António Egas Moniz devised the operation and received the Nobel prize in medicine for it in 1949, which has never been rescinded. Freeman took it up in America and developed a version performed through the eye socket in under ten minutes, without a surgical theatre. He carried it out around three and a half thousand times, sometimes on children, often on patients whose diagnosis would not now warrant any surgery at all. He kept no adequate outcome data. The operation was abandoned in the 1950s as antipsychotic drugs arrived.

WHAT THE ERROR WAS MADE OF

A procedure judged by the convenience of the ward rather than the state of the patient. Lobotomised patients were calmer, easier to manage, and could sometimes go home, and in an era of vast overcrowded asylums with nothing else to offer, that was recorded as improvement. Nobody was systematically asking what had been taken away, because the person best placed to report it was the person who had been operated on. When the measure of success is chosen by whoever is inconvenienced by the problem, the treatment will succeed. Ask who chose the measure.

046

“We accept an interest in people’s health as a basic responsibility, paramount to every other consideration in our business. There is no proof that cigarette smoking is one of the causes of lung cancer.”

The Tobacco Industry Research Committee, 1954

WHAT HAPPENED

It held for thirty years. The statement ran as a full-page advertisement in more than four hundred American newspapers. It answered studies published in 1950 by Doll and Hill in Britain and by Wynder and Graham in America. The committee it announced funded research for decades. The American Surgeon General’s report came in 1964, and the industry maintained a public position of unresolved scientific controversy for thirty years after that. Internal documents released in litigation in the 1990s showed that company scientists had reached the opposite conclusion privately.

WHAT THE ERROR WAS MADE OF

Nothing, in the sense of this book. This is the counterfeit, and it is here so that you can compare it with the genuine article on either side. It has the same surface as an honest scientific caution, the same appeal to more research, the same insistence on proof rather than evidence. The difference is invisible from outside and total: an honest doubt is looking for the answer, and this one was built to postpone it.

047

Thalidomide is so free of toxicity that it can safely be given to pregnant women for morning sickness.

Chemie Grünenthal, 1957

WHAT HAPPENED

Ten thousand children. It was sold over the counter in dozens of countries and was promoted as exceptionally safe, largely because it was hard to overdose on. Somewhere over ten thousand children were born with severe limb malformations, and many more pregnancies ended. It was withdrawn in late 1961 after the Australian obstetrician William McBride and the German paediatrician Widukind Lenz independently identified it. The United States was largely spared because Frances Kelsey at the FDA refused to approve it while she waited for data she had asked for and not been given.

WHAT THE ERROR WAS MADE OF

A safety claim resting on the absence of a particular harm. The drug really was strikingly non-toxic to the person who took it, and that was the tested thing. Nobody had tested what it did to a developing embryo, because before this the assumption in the field was that the placenta protected the foetus from most compounds. The general principle is that *safe* is never a property of a substance on its own. It is a property of a substance, a dose, and a body, and the body that had not been considered here was the one that mattered. Safe for whom, and at what dose.

048

A pandemic on the scale of 1918 is likely this winter, and the entire American population should be vaccinated.

David Sencer, 1976

WHAT HAPPENED

The pandemic never came. A soldier at Fort Dix died of a swine influenza strain and several others were infected. Sencer, who ran the Centers for

Disease Control, pressed for a national immunisation programme, and President Ford announced it. Around forty-five million Americans were vaccinated. The pandemic never came; the virus did not spread beyond the base. The campaign was suspended after an apparent rise in Guillain-Barré syndrome among those vaccinated, and Sencer lost his job.

WHAT THE ERROR WAS MADE OF

A decision under genuine uncertainty, which is the reason it is in this chapter rather than being simply a mistake. The chance of a 1918-scale event was small and the cost if it happened was immense, and on those numbers acting was defensible. What was not defensible was announcing it with the confidence of a forecast rather than the honesty of a bet. If Sencer had said publicly that this was insurance against an unlikely catastrophe, the programme's failure would have been a cost of doing the sensible thing. Told as a prediction, it read as a lie, and the damage to public trust in vaccination outlasted everyone involved. Say bet when you mean bet.

049

The proposition that a bacterium causes peptic ulcers ranks in the bottom tenth of the abstracts submitted, and is not worth a place on the programme.

The Gastroenterological Society of Australia, 1983

WHAT HAPPENED

Two years, and a flask of bacteria. Barry Marshall and Robin Warren had cultured Helicobacter pylori from the stomachs of ulcer patients. The consensus was that ulcers were caused by stress and acid, that nothing could live in stomach acid, and that treatment meant acid suppression for life or surgery. Their abstract was rejected. In 1984 Marshall drank a flask of the culture, gave himself gastritis, and demonstrated the point on himself. Ulcers are now cured with a fortnight of antibiotics. The two shared the Nobel prize in 2005.

WHAT THE ERROR WAS MADE OF

A background assumption nobody had checked because it was too obvious to check. Everything followed from *nothing lives in the stomach*, which was not a finding but an inference. Pathologists had been seeing spiral bacteria in gastric biopsies for decades and recording them as contamination. The most dangerous belief in any field is not the contested one. It is the one so uncontroversial that no experiment has ever been pointed at it. Ask what nobody has tested.

050

Vioxx is a safer anti-inflammatory, gentler on the stomach, appropriate for long-term use in arthritis.

Merck, 1999

WHAT HAPPENED

It was withdrawn in 2004. It was prescribed to something like eighty million people. The VIGOR trial in 2000 was designed to show the gastrointestinal advantage. It also showed a much higher rate of heart attacks in the Vioxx arm than in the comparator. The published interpretation put this down to a protective effect of the older drug, and not to a harmful effect of the new one. The company withdrew Vioxx in 2004 after a further trial confirmed the cardiovascular risk. Estimates of the excess events run into the tens of thousands.

WHAT THE ERROR WAS MADE OF

An unexpected finding explained away by the hypothesis it threatened. The alternative reading was available to everyone from the moment the data came in, and it was less comfortable and turned out to be correct. It is worth being precise, because the failure is not exotic. A trial designed to answer one question produced an answer to a different one, and the different one was assessed by people who had already committed. What protects against this is not integrity, which most of them had. It is having the awkward result adjudicated by somebody with nothing at stake. Give the awkward result to a stranger.

CHAPTER 6

Safe, According to the Model

An engineer's confidence is different from a scientist's, because it is written down as a number and somebody's life is resting on it.

That ought to make it more reliable, and mostly it does. The buildings stand up. What this chapter collects is the small proportion of cases where the number was produced correctly by a method that had quietly stopped describing the thing being built. The arithmetic looks identical either way. A calculation cannot tell you that it is answering the wrong question.

Three failures recur. The first is the untested extrapolation. A design method that has worked for twenty structures is applied to one that is longer, thinner, faster or deeper than any of them. The range over which it was validated goes unmentioned, because everyone has forgotten there was one. The second is the missing failure mode, where every load anybody thought of was analysed properly and the one that killed it was not on the list. The third is organisational, and it is the worst: the number was right, somebody said so, and the structure was built anyway.

Notice how many of these had a warning. Not a vague misgiving afterwards, but a specific person saying a specific thing beforehand, in writing, and being overruled or ignored. That pattern is so consistent it is nearly a law, and it is the strongest practical argument in this book. If you want to know whether your project is about to fail, the useful question is not whether you are confident. It is who has already told you it will, and what happened to them. The shape is the same in a warship of 1628 and a submersible of 2023: somebody measured the thing, somebody said out loud what the measurement meant, and the decision was taken by a person for whom stopping had become more expensive than continuing.

051

The ship is stable enough. Sail her.

Gustavus Adolphus, 1628

WHAT HAPPENED

She sank in the harbour. The Vasa was the most powerful warship in the Baltic, built in haste for a war, with a second gun deck added at the king's request while she was on the stocks. Before launch the shipyard ran a stability test in which thirty men ran from side to side across the deck; it was stopped after three passes because the ship was rolling dangerously. She sailed on 10 August 1628, caught a gust in Stockholm harbour, heeled over, took water through the open lower gunports and sank about 1,300 metres into her maiden voyage. The inquiry found nobody at fault. She was raised in 1961 and is intact.

WHAT THE ERROR WAS MADE OF

A test that was performed, understood and then not allowed to mean anything. This is the purest version of the pattern in this chapter: nobody lacked the information. The men on the deck knew, and so did the officer who stopped the test. The knowledge did not travel upwards, because the man it had to reach had already decided and was at war. Every subsequent entry here is a more sophisticated version of the same thirty men running from side to side.

052

"The captain can, by simply moving an electric switch, instantly close the doors throughout, practically making the vessel unsinkable."

The Shipbuilder, 1911

WHAT HAPPENED

One year. The Titanic's hull was divided into sixteen watertight compartments and could float with any two of them flooded, or with the first four. The iceberg opened six. She sank on 15 April 1912 with about 1,500 people aboard, having carried lifeboats for roughly half those on board, which was more than the regulations of the day required.

WHAT THE ERROR WAS MADE OF

A design case treated as a guarantee. The compartment system was real, it was well engineered, and the phrase in the trade press was hedged with the word *practically*, which everybody dropped. The failure is worth stating plainly, because it repeats. The ship was designed against the worst accident anyone had seen, which was a collision at a bulkhead. The sea is not obliged to stay inside the range of your experience. The lifeboats are the more damning half. They were adequate for the ship in the model and not for the ship that existed.

053

The deflection theory permits a far lighter and shallower deck than earlier practice, at no cost in safety.

Leon Moisseiff, 1940

WHAT HAPPENED

Four months. The Tacoma Narrows Bridge opened in July 1940 with a deck eight feet deep on a span of over eight hundred metres, the most slender suspension bridge ever built. It moved in light winds from the first day and was nicknamed Galloping Gertie by the people who drove over it. On 7

November, in a wind of about forty miles an hour, well below its design load, it went into a twisting oscillation and tore itself apart. The film of it is the most-watched engineering failure in history.

WHAT THE ERROR WAS MADE OF

A method extrapolated past the conditions that validated it. Moisseiff's theory was sound and had produced excellent bridges; it treated wind as a static horizontal force, which is adequate for a stiff deck and catastrophic for a flexible one. Aerodynamic flutter was not a phenomenon suspension bridge engineers had needed to think about, because no previous deck had been light enough to flutter. Every design method has a range of validity, and the range is invisible from inside it until something leaves it.

054

The pressurised cabin is amply strong. The airframe is good for tens of thousands of cycles.

de Havilland, 1952

WHAT HAPPENED

Three broke up in the air. The Comet was the first jet airliner and a genuine triumph, in service in May 1952 and a year ahead of anything else in the world. Three broke up in flight within twenty-six months. The investigation submerged an entire fuselage in a water tank and pressurised it thousands of times. It found metal fatigue starting at the corners of cutouts in the skin, where the stress concentrated far more than the calculations assumed. The design was corrected, the type flew safely for decades afterwards, and Boeing and Douglas took the market.

WHAT THE ERROR WAS MADE OF

A material behaviour that had not mattered before and did now. Fatigue was known. What was new was a pressurised cabin cycled twice a day behind a thin skin at high altitude, and the safety factor that covered every previous aircraft did not cover it. The reason this entry is here rather than in a chapter about ordinary bad luck is the water tank: the test that found the answer was not exotic, and it existed. It was simply not thought necessary until after the crashes.

055

The walkway connection as fabricated is equivalent to the design.

The Hyatt Regency engineers, 1979

WHAT HAPPENED

Two years. Two suspended walkways in the atrium of the Kansas City hotel were to hang from single continuous rods. During fabrication the detail was changed to two shorter rods, with the upper walkway hanging from the lower one's box beam, because a continuous threaded rod is awkward to make and install. The change doubled the load on that connection. On 17 July 1981, during a crowded tea dance, both walkways came down. A hundred and fourteen people died. It remains one of the worst structural failures in American history, and both engineers lost their licences.

WHAT THE ERROR WAS MADE OF

A change nobody costed because it looked like a detail. The original design was already marginal, which is the part usually left out, and the modification turned marginal into fatal. What makes it the standard teaching case is how ordinary the failure is. A fabricator proposed a simpler way to build a joint. A shop drawing came back and somebody approved it under time pressure without redoing the arithmetic, and the arithmetic was five minutes of work. Almost nothing here required expertise. It required somebody to treat a small change as a change.

056

“A reactor of this type could be installed on Red Square. It is no more dangerous than a samovar.”

attributed to Anatoly Aleksandrov, 1980

WHAT HAPPENED

Unit 4 exploded. Aleksandrov was president of the Soviet Academy of Sciences and one of the fathers of the RBMK design. The reactor had a positive void coefficient, meaning that in certain conditions a loss of cooling increases the reaction rather than damping it, and control rods that briefly added reactivity as they entered. Both were known to the designers and neither was in the operating manuals at Chernobyl. Unit 4 exploded on 26 April 1986 during a test of the turbine at low power.

WHAT THE ERROR WAS MADE OF

A hazard known at the top and absent at the bottom. This is what happens where admitting a defect is politically costly. The information does not disappear. It gets classified, and the people who need it most are cleared least. The operators at Chernobyl did the wrong thing that night, and there is no version of the story in which they could have known why it was wrong. A safety culture is not a set of attitudes. It is a question about who is allowed to know what.

057

The probability of a catastrophic failure is about one in a hundred thousand flights.

NASA management, 1986

WHAT HAPPENED

Seventy-three seconds. Challenger broke up seventy-three seconds after launch on 28 January 1986, on the twenty-fifth shuttle flight. The night before, engineers at Morton Thiokol had recommended against launching. The O-ring seals in the solid rocket boosters had never been tested near the forecast temperature and had shown erosion on earlier cold flights. They were asked to reconsider as managers rather than as engineers, and the recommendation was reversed. Richard Feynman sat on the commission afterwards. He found that working engineers put the odds of loss at around one in a hundred while management used a figure a thousand times better, and he asked, in the report's appendix, what the basis for it was.

WHAT THE ERROR WAS MADE OF

Evidence of a problem reclassified as evidence of margin. The O-rings had been eroding for years without a loss. That record was read as proof that erosion was survivable, and not as a warning that the seal was not doing its job. Once a deviation has occurred a few times with no consequence it stops being a deviation and becomes the expected condition, and the standard quietly moves. Feynman's closing line is the one to keep: for a successful technology, reality must take precedence over public relations, for nature cannot be fooled.

058

The design basis tsunami for the site is 5.7 metres. Higher estimates are not sufficiently established to require action.

TEPCO, 2008

WHAT HAPPENED

The wave was fifteen metres. An internal study in 2008 concluded that a tsunami of over fifteen metres was possible at Fukushima Daiichi on the basis of historical events, including one in the year 869. It was treated as provisional and passed for further study rather than acted on. On 11 March 2011 the wave reached roughly fourteen to fifteen metres at the site, flooded the emergency generators in the basements, and three reactors melted down. The Diet's independent commission called the accident profoundly man-made.

WHAT THE ERROR WAS MADE OF

A conclusion nobody had to reject because it could be kept pending. The finding was made by the company's own people and never overturned. It was held in the queue, because acting on it meant an enormous expense against an event that had not happened in a thousand years. This is not ignorance, and it is far more common. An organisation never has to decide against an inconvenient finding, because indefinite study is available and looks responsible from every angle.

059

The negative pressure test result is anomalous but explainable. The well is secure.

BP and Transocean, 20 April 2010

WHAT HAPPENED

The well blew out that evening. A test to confirm the Macondo well was sealed produced pressure readings that should not have been possible if the seal were good. The crew debated it for an hour and settled on an explanation involving a bladder effect in the drilling fluid, a phenomenon with no real basis. They proceeded. The well blew out that evening, eleven men died, the rig sank, and around four million barrels of oil went into the Gulf of Mexico over the following three months.

WHAT THE ERROR WAS MADE OF

A rescuing explanation invented at the point of decision. The reading was unambiguous and everyone saw it. One interpretation meant stopping, which was expensive and embarrassing forty days behind schedule. The other meant continuing. So the group looked for a reason the instrument might be lying, and found one. A hypothesis produced only in order to permit the action you already wanted is not a hypothesis, and the moment to notice this is when it is being generated, not afterwards.

060

"At some point safety is just pure waste. I mean, if you just want to be safe, don't get out of bed, don't get in your car, don't do anything."

Stockton Rush, 2019

WHAT HAPPENED

Four years. Rush's company took paying passengers to the wreck of the Titanic in a submersible with a carbon fibre hull. The material had no service history at that depth and is known to degrade in ways that are hard to inspect. He declined classification by any marine society, arguing publicly that certification stifles innovation. Employees and outside experts wrote to him warning of catastrophic risk; one was dismissed. The vessel imploded in June 2023, killing him and four others.

WHAT THE ERROR WAS MADE OF

A true general observation used to dismiss a specific one. It is perfectly correct that safety has diminishing returns and that a fully risk-averse life is not a life. None of it says anything about whether a particular pressure hull will hold at four thousand metres. The move is a common one and worth being able to name: a philosophical point about risk in general, deployed to avoid an engineering point about this object. The warnings he received were not about the value of caution. They were about the hull.

III

MONEY

CHAPTER 7

A Permanently High Plateau

Financial forecasting is the only field in this book where being wrong is the normal condition and everybody knows it. That should make it humble. It does the opposite, and the reason is worth understanding before you read a single entry.

A forecast about flight is checked by an aeroplane. A forecast about the market is checked by other people's forecasts, and in the short run the market does whatever the consensus does, so the consensus is correct until abruptly it is not. Anybody who dissents is wrong for years first, and being wrong for years is fatal to a career even when it ends in being right. The incentives do not reward accuracy. They reward being wrong at the same time as everybody else, which is survivable, over being right alone, which is not.

Then there is the position of the speaker. Several people in this chapter were paid by an industry to comment on that industry, and at least one has said so himself since, in as many words. That is not fraud and it does not feel like distortion from inside. It is that a man whose salary depends on a market does not have the same relationship to bad news about it as you do.

The last thing to notice is the vocabulary. *Contained. Froth. Transitory. Localised.* Every one of these is a word that acknowledges a problem while denying it can travel, and the whole history of financial crisis is the history of things travelling. When you see a forecast that concedes the fact and disputes the transmission, mark it. That is the exact place where this chapter's errors live. It is worth holding on to the fact that every person quoted here had access to the same numbers as the people who got it right, and in several cases had produced those numbers himself, which is why none of these are failures of information.

061

"Stock prices have reached what looks like a permanently high plateau."

Irving Fisher, 1929

WHAT HAPPENED

Eight days. He said it on 16 October. The market broke eight days later and lost about eighty-nine per cent of its value over the next three years. Fisher was the leading American economist of his generation and one of the founders of modern monetary theory. His work on debt deflation, written afterwards to explain what had happened, is still read. He was also personally invested and lost everything, including his house, which Yale bought and rented back to him for the rest of his life.

WHAT THE ERROR WAS MADE OF

A new theory of value arriving at the top of a boom. Fisher's argument was that prohibition and scientific management had permanently raised the productivity of American industry, so the higher earnings were real and the prices justified. It is a serious argument, and some of it was true. The trouble with a story that explains why the old valuation rules no longer apply is that one always appears near a peak, because that is precisely when people need one. The story arrives with the price.

062

"The fundamental business of the country, that is the production and distribution of commodities, is on a sound and prosperous basis."

Herbert Hoover, 25 October 1929

WHAT HAPPENED

He said it the day after the first great break. Industrial production fell by about half over the next three years, unemployment reached roughly a quarter of the workforce, and thousands of banks failed. Hoover was not a stupid man and he was not idle; he did more than any previous president had to intervene, which was still not remotely enough.

WHAT THE ERROR WAS MADE OF

A statement whose purpose was to steady the room, said in the voice of an assessment. There is a real dilemma here and it is not hypocrisy. A president who says the truth about a bank run makes the run worse, and every official facing a panic has to weigh accuracy against the effect of saying it. The cost comes later. Once the reassurance has failed, nothing that office says about the economy is believed for a decade, and that mattered more than any single day's confidence.

063

The Harvard Economic Society's index indicates that a severe depression is outside the range of probability. A recovery in business is to be expected shortly.

The Harvard Economic Society, 1929

WHAT HAPPENED

They called the trough again and again. The Society was the most respected forecasting operation in America, run by academics using a statistical barometer of leading indicators. Through 1930 and 1931 it repeatedly announced that the trough had been reached. It was wrong at every turn, lost its subscribers, and was wound up in 1931.

WHAT THE ERROR WAS MADE OF

A model fitted to a range of history that did not contain the event. The barometer had been built from the business cycles of the previous thirty years, none of which was a depression. It could describe a recession accurately and had no vocabulary at all for what was happening. This is the most reliable failure in quantitative forecasting and better statistics will not fix it. A model can only recognise a kind of event that occurred in the data it was fitted to. The data had never seen one.

064

The models put a loss of this size at many standard deviations, an event not expected within the lifetime of the universe.

Long-Term Capital Management, 1998

WHAT HAPPENED

Four years. The fund was run by John Meriwether, with Myron Scholes and Robert Merton on the board. Both had won the Nobel prize in economics the previous year, for the mathematics of option pricing. It had returned about forty per cent a year on enormous leverage. Russia defaulted in August 1998, correlations across every unrelated market went to one, and the fund lost most of its capital in weeks. The Federal Reserve Bank of New York organised fourteen banks to take it over, because its counterparty exposures were large enough to threaten the system.

WHAT THE ERROR WAS MADE OF

A distribution assumed rather than measured. The models treated price moves as normally distributed and the positions as independent, and both assumptions hold comfortably in ordinary conditions and fail together in exactly the conditions that matter. The deeper point is the one their own arithmetic should have made: if your model says an event cannot happen and the event happens, the correct inference is not that you were unlucky. It is that the model was the wrong one, and it always was. Ten sigma means check the model.

065

There is no national housing bubble. The boom will not bust.

David Lereah, 2005

WHAT HAPPENED

Prices fell by a third. Lereah was chief economist of the National Association of Realtors. He published a book in 2005 urging readers not to miss the real estate boom, and reissued it the following year with a title promising the boom would not bust. American house prices peaked in 2006 and fell by about a third nationally, more in the states he was most positive about. He left the association in 2007 and has since said, with some grace, that he was a shill for an industry and that he should have seen it.

WHAT THE ERROR WAS MADE OF

A conflict of interest operating at the level of what looked worth investigating. He was not, as far as anyone can show, saying things he believed to be false. He was employed by an industry to explain that industry to the public. In that position the arguments for a rise get examined thoroughly and the arguments for a fall get answered quickly. Ask who pays the forecaster before you weigh the forecast, not because they are lying, but because it tells you which half of the evidence has been checked hard. Ask who pays.

066

"We do have characteristics of bubbles in certain areas, but not, as best I can judge, nationwide."

Alan Greenspan, 2005

WHAT HAPPENED

He recanted in public, later. Greenspan had run the Federal Reserve since 1987 and was treated in the press as an oracle. His position was that housing markets are local, that there was froth in some of them, and that a simultaneous national decline had no precedent. Prices fell nationally from 2006, the mortgage securities built on them failed, and the banking system came close to collapse in 2008. In congressional testimony in October 2008 he said he had found a flaw in the ideology he had held for forty years. It is a more honest sentence than most people in this book ever produced.

WHAT THE ERROR WAS MADE OF

Absence of precedent read as evidence of impossibility. He was right that there had never been a nationwide fall in American house prices in the modern data. The data began in the 1940s. What made a national fall possible this time was a national market in mortgage securities that had not existed before. The innovation that made the crash possible was also the reason the historical record looked reassuring. When somebody tells you a thing has never happened, the question is whether the world that prevented it is still the world you are in. Ask what has changed since.

067

These senior tranches merit a triple-A rating, equivalent in default risk to a sovereign bond.

Moody's and Standard and Poor's, 2006

WHAT HAPPENED

They were downgraded ten notches. Trillions of dollars of securities built from subprime mortgages were rated at the highest grade available, which is what allowed pension funds and banks to hold them at all. Large numbers were downgraded by ten or more notches within eighteen months, some to default. The agencies were paid by the issuers of the securities they were rating.

WHAT THE ERROR WAS MADE OF

A correlation assumption at the centre of the arithmetic, chosen because it made the model tractable. Pooling mortgages removes risk only if the mortgages fail independently, and in a national downturn they do not. There was no secret about this; it is the first thing anyone says about pooling. What kept the wrong number in the model was that the right number destroyed the product. No individual had to be corrupt for an institution paid by issuers to prefer the assumption that let it issue.

068

"We believe the effect of the troubles in the subprime sector on the broader housing market will likely be limited, and we do not expect significant spillovers to the rest of the economy or to the financial system."

Ben Bernanke, May 2007

WHAT HAPPENED

Sixteen months. Lehman Brothers failed, the money markets froze, and the Federal Reserve, which Bernanke chaired, undertook the largest emergency intervention in its history. To his considerable credit he then ran that intervention with real skill. He was the leading academic historian of the Great Depression and knew exactly what a central bank must not do.

WHAT THE ERROR WAS MADE OF

Sizing a problem by the assets rather than by the plumbing. The subprime mortgages themselves were a small slice of American credit, and the arithmetic on that was correct and was widely quoted. What travelled was not the losses; it was the discovery that nobody could tell which institution held them, which turned a manageable loss into a general refusal to lend to anybody. Contagion in a financial system runs through uncertainty about counterparties, and no calculation of the size of the underlying asset will ever capture it. Size the plumbing, not the pipe.

069

"Bear Stearns is fine. Do not take your money out."

Jim Cramer, 11 March 2008

WHAT HAPPENED

Five days. Bear Stearns was sold to JPMorgan Chase in a Federal Reserve-brokered rescue at two dollars a share, against a price of about a hundred and seventy the previous year. Cramer has argued since, with some justice, that the viewer was asking whether his brokerage account was safe and not whether the stock was. A Bear Stearns account holder did get their money out.

WHAT THE ERROR WAS MADE OF

A precise answer delivered in a format that cannot carry precision. Whatever he meant, what a viewer heard on a television programme, shouted, with sound effects, was that the firm was sound. The lesson is not about him. It is that the medium determines what a message can contain, and a qualification that requires the listener to have caught the exact question does not survive the format it was said in. If a distinction is doing all the work, do not make the statement somewhere the distinction will be lost.

070

The current increase in inflation is transitory, driven by pandemic reopening effects that will fade.

Jerome Powell, 2021

WHAT HAPPENED

Nine per cent. American consumer price inflation, running at about one and a half per cent when the word was first used, reached nine per cent in June 2022. The Federal Reserve retired the word *transitory* in November 2021 and then raised interest rates faster than at any time since the early 1980s. Inflation did come down without the recession most forecasters expected, so the underlying judgement about supply chains was partly vindicated and the timing was badly wrong.

WHAT THE ERROR WAS MADE OF

A word without a time limit doing the work of a forecast. *Transitory* has no duration in it, so it could not be wrong on any particular date, and by the time it was obviously wrong the moment to act had gone. It also functioned as a reason not to act, which is the most useful property a vague word can have inside an institution that does not want to move. Any forecast that cannot be falsified on a stated date is not a forecast. It is a mood, and it will be defended long past the point where a number would have been abandoned. Put a date on it.

CHAPTER 8

The Deal They Turned Down

This chapter is the cruellest in the book, and it is also the one most likely to make you unfairly pleased with yourself, so it comes with a warning attached.

Every entry is a decision that looks insane now and was reasonable at the time. That is not a defence offered out of politeness. It is the actual finding. Read these as stories about idiots and you will learn nothing you can use. You are not going to be offered Google for a million dollars, and if you were you would say no for the same reasons they did.

The reasons are worth naming in advance. A large company evaluates an opportunity against its existing business, which is the only yardstick it has, and by that yardstick every important new thing is small, unprofitable and a distraction. It is not that executives fail to see the new object. They see it clearly and correctly conclude that it is worth almost nothing compared with what they already own. They are right about the present and the comparison is the mistake.

The second reason is that the value in most of these deals was not in the asset. It was in what the buyer would then have had to do. No valuation can price an organisation's willingness to destroy its own profitable business in order to build a worse one. Kodak understood digital photography better than anyone alive. That knowledge was the problem, because it told them precisely how much money they would lose.

Finally, the part that gets left out. For every declined deal in this chapter there are a hundred that were declined correctly, and a hundred more that were accepted and destroyed the acquirer. We remember these because of what the thing became. Nobody keeps a list of the offers that would have been disasters. The right way to read the chapter is as a series of correct answers to the wrong question, because in nearly every case the executive was asked what the thing was worth to the company as it stood, and answered accurately, and the company as it stood was the part that was about to end.

071

The telephone is a scientific curiosity with no commercial application. Decline the offer.

attributed to Western Union, 1876

WHAT HAPPENED

They declined. Bell's backers offered the telephone patents to Western Union, then the dominant communications company in America, for a hundred thousand dollars. Western Union declined. Within four years it was in litigation with the Bell company, which it lost, and by 1900 Bell was worth many times what Western Union had ever been. The famous internal memo about a device with too many shortcomings is quoted everywhere and has never been produced; what is solid is the decision itself.

WHAT THE ERROR WAS MADE OF

Judging a new medium as a worse version of the old one. Western Union's business was messages between cities, delivered by trained operators to offices. A telephone in 1876 carried a poor voice a few miles between two fixed points. As a telegraph it was hopeless. That comparison is accurate and it is the wrong comparison, because the telephone was not competing for telegrams. It was creating a market of conversations that had never existed, and there is no way to see that market by looking harder at the one you have.

072

“Guitar groups are on the way out. The Beatles have no future in show business.”

attributed to Decca Records, 1962

WHAT HAPPENED

They signed somebody else. Decca auditioned the group on the first of January 1962 and turned them down, signing Brian Poole and the Tremeloes instead, partly because they were from London and cheaper to work with. Parlophone signed the Beatles later that year. The line is Brian Epstein’s account of what Dick Rowe told him, and Rowe denied the wording for the rest of his life. He also signed the Rolling Stones the following year on George Harrison’s recommendation, which is the part usually left out.

WHAT THE ERROR WAS MADE OF

A judgement of taste presented as a judgement of trend. The audition was poor by all accounts, the material was mostly covers, and a professional listening that day would have heard a competent northern club act. Rowe’s actual mistake, if he made one, was to convert that into a claim about the direction of popular music, which is a different and much harder thing to know. He was allowed to be wrong about one band. What sank him was the sentence about the category.

073

It is a cute demonstration, but nobody wants to look at photographs on a television, and we should not talk about it.

Kodak, 1975

WHAT HAPPENED

They put it in a drawer. Steven Sasson, a young engineer at Kodak, built the first self-contained digital camera: a toaster-sized device recording to cassette tape at ten thousand pixels, taking twenty-three seconds a picture. He demonstrated it to management, who by his account found it charming and told him not to tell anyone. Kodak went on to invest heavily in digital, patent a great deal of it, and earn substantial licence income, and it still could not bring itself to attack its own film business. It filed for bankruptcy in 2012.

WHAT THE ERROR WAS MADE OF

Not blindness, which is the version everybody tells, but arithmetic. Kodak's profit came from film, paper and processing, sold repeatedly to the same customer for the life of a camera. Digital photography removed all three at once. The internal analysis was correct in every particular, including its estimate of when digital would become good enough. The conclusion it supported was to defend the profitable business as long as possible. There is no obvious way for a public company to choose voluntarily to make one-tenth as much money, and that, not stupidity, is what the word disruption actually means.

074

A tenth of this company is not worth the liability that comes with it.

Ron Wayne, 1976

WHAT HAPPENED

He took the eight hundred dollars. Wayne was the third founder of Apple, drew the first logo, wrote the original partnership agreement, and held ten per cent. Twelve days in, he relinquished his stake for eight hundred dollars, later accepting a further fifteen hundred to formalise it. He was the only partner with personal assets. In a general partnership he would have been liable for the company's debts, which at that moment were real. The revenues were not.

WHAT THE ERROR WAS MADE OF

Almost nothing, which is why it is here. Wayne was forty-one with things to lose, next to two men in their twenties with nothing, and he made a defensible decision about risk given his own position. He has said many times since that he does not regret it and would probably have been squeezed out anyway. The entry is a corrective to the rest of the chapter. An outcome that turns out enormous does not make the decision that avoided it stupid. Judging a choice by a result the chooser could not see is the standard way to learn the wrong lesson.

075

A non-exclusive licence is acceptable. The value is in the hardware.

IBM, 1980

WHAT HAPPENED

The clones arrived. IBM needed an operating system for its personal computer in a hurry and licensed one from Microsoft, allowing Microsoft to sell the same product to anyone else. Compaq reverse-engineered the IBM firmware in 1982, the clone industry followed, and every clone needed the operating system. IBM sold its personal computer division in 2005. Microsoft became for a period the most valuable company on earth.

WHAT THE ERROR WAS MADE OF

Locating the value where it had always been. In IBM's world hardware was the product and software was the thing you gave away to sell it, and that had been true for thirty years and enormously profitable. What nobody at the table could see was that the standard would become the asset once the hardware was commoditised, because hardware had never been commoditised before. The general form is worth carrying: in any deal, ask which part of this will still be scarce when the rest is easy.

076

A search engine is a feature, not a business, and this one is not worth a million dollars.

Excite, 1999

WHAT HAPPENED

They said no more than once. Larry Page and Sergey Brin, wanting to get back to their doctorates, tried to sell Google to the portal companies for around a million dollars and were turned down more than once. Excite's management concluded that a search engine which sent users away quickly was the opposite of what a portal needed. Excite was bought, went through bankruptcy in 2001, and disappeared.

WHAT THE ERROR WAS MADE OF

A strategy correctly applied to the wrong future. The portal business model was to keep people on the page. Excite's objection to Google, that it was too good at getting people to leave, was a correct description of the product. What they could not price was that being the place people start is worth more than being the place they stay. Nobody in 1999 knew what an auction of search intent would fetch, because that market had not been invented.

077

Netflix is a very small niche business. Fifty million dollars is far too much.

Blockbuster, 2000

WHAT HAPPENED

They laughed. Reed Hastings and Marc Randolph flew to Dallas and offered to sell Netflix, then a mail-order DVD service losing money, to Blockbuster for fifty million dollars, proposing to run its online arm. By Randolph's account they were barely able to keep a straight face in the room. Blockbuster filed for bankruptcy in 2010. Netflix is worth several hundred times the price it asked.

WHAT THE ERROR WAS MADE OF

A profit centre that could not be criticised. Blockbuster's earnings depended substantially on late fees, which customers hated and which the Netflix model abolished. Any serious internal assessment of Netflix was therefore an argument for destroying a line of revenue that everybody had been rewarded for growing. The decisive fact about an incumbent is rarely what it fails to understand. It is which sentence cannot be said out loud in its own boardroom.

078

Running an online bookshop is not our competence. Amazon should run it for us.

Borders, 2001

WHAT HAPPENED

Amazon kept the customer. Borders handed its entire online operation to Amazon, which ran borders.com and kept the customer relationship, the data and the margin, while Borders got a fee and concentrated on its stores. The arrangement ran until 2008. By then Amazon knew what every Borders customer bought online and Borders had built nothing. It liquidated in 2011.

WHAT THE ERROR WAS MADE OF

Outsourcing a cost and giving away an asset. The reasoning was sound as far as it went: Borders was bad at the web, Amazon was superb at it, and specialisation is usually right. What the analysis missed is that the thing being outsourced was not a warehouse function but the relationship with the customer in the channel that was about to become the whole market. Before you contract out an activity, ask whether it is a cost centre or the place your business will live in ten years, because the second one is never worth a fee.

079

"It's kind of one more entrant into an already very busy space. But in terms of a sea change for BlackBerry, I would think that's overstating it."

Jim Balsillie, 2007

WHAT HAPPENED

Twenty per cent to under one. Balsillie was co-chief executive of Research In Motion, whose BlackBerry then dominated corporate email and was the object of genuine addiction among the people who carried one. RIM's share of the smartphone market went from about twenty per cent to under one per cent within six years. The company survives as a software and security business and has not made a phone since 2016.

WHAT THE ERROR WAS MADE OF

Defending a moat that was about to become irrelevant rather than being crossed. RIM's advantages were real: superb battery life, a keyboard people typed on at speed, and an enormously efficient network protocol that corporate IT departments and carriers loved. Every one of those was an answer to a constraint, and within four years batteries, screens and bandwidth had all improved enough that the constraints were gone. An advantage built on somebody else's limitation expires when that limitation does, and the expiry date is not on the balance sheet.

080

Thirty-one dollars a share undervalues the company. Reject it.

Yahoo, 2008

WHAT HAPPENED

They held out for more. Microsoft offered about forty-four billion dollars for Yahoo in early 2008. Jerry Yang and the board held out for more, the offer was withdrawn, and the shares fell by more than half within the year. Yahoo's core internet business was sold to Verizon in 2016 for around four and a half billion, roughly a tenth of the rejected price.

WHAT THE ERROR WAS MADE OF

An anchor set by the company's own history rather than by the offer in front of it. Yahoo had been worth far more in 2000, its management remembered that number, and thirty-one dollars felt like an insult measured against it. The past price of your own asset carries no information about its future price. It is the most powerful distorting fact in any negotiation, which is why the other side will mention it. The only relevant question is what it will be worth later, and the offer itself is usually the best available evidence about that.

CHAPTER 9

The Thing That Only Goes Up

A bubble is the one error in this book that requires other people. Everything else here could have been said by a man alone in a room. This cannot, because the whole mechanism is that the price is the argument and the argument is the price.

That makes it uniquely hard to escape by thinking. You can be entirely correct about the value of an asset, say so, and watch it triple. Every day it triples is a day the market tells you, and everyone watching, that you were wrong. The economist's version of this is that markets can stay irrational longer than you can stay solvent. The practical version is that being early is indistinguishable from being wrong, both to your clients and, after a while, to yourself.

Two things are worth having clear before the entries. The first is that most people in a bubble know it is a bubble. That is not a secret discovered afterwards; contemporary accounts are full of people saying so while buying, on the theory that they will get out first. The mania is not usually a failure to see what is happening. It is a bet on timing, made by everybody at once, and the timing is the only part that cannot be right for everybody.

The second is that a bubble almost always forms around something real. Railways did change England, the internet did change everything, houses are worth living in. The story is true. That is what makes the price possible, and it is why *but the technology is genuinely important* is the least useful sentence anybody can offer you at the top. It is always true at the top. It was true in 1845 and 1999, and the shares still fell ninety per cent.

Almost nobody here was ignorant: a master of the mint, a monetary theorist, the author of the standard work on financial mania and two economists writing from a research institute all lost their footing in a market they had studied.

081

A single bulb is worth a house on the Herengracht, and the price will hold, because the supply cannot be increased.

The Amsterdam tulip trade, 1637

WHAT HAPPENED

It stopped in a week. In the winter of 1636 the trade in tulip contracts, mostly conducted in taverns for delivery in spring, rose to extraordinary levels. In the first week of February 1637 the bids stopped. Prices fell by well over ninety per cent. The historian Anne Goldgar's work in the Dutch archives suggests the wider economic damage was far smaller than the legend claims and that few were ruined. The story of a nation destroyed by flowers was written later, by moralists who found it useful.

WHAT THE ERROR WAS MADE OF

A scarcity argument that ignored biology and the contract. Tulips propagate, slowly but reliably, so the supply was not fixed. More to the point, what was being traded by the end was not bulbs but promises to buy bulbs, and promises can be created without limit. Whenever a price is defended by the impossibility of making more of the thing, check whether the thing being traded is the thing or a claim on it. The claims are never scarce. Claims are printed, not grown.

082

The company's American trade will service the debt of France, and the shares are cheap at any price.

John Law, 1720

WHAT HAPPENED

It fell apart within a year. Law, a Scottish gambler and monetary theorist, persuaded the French regent to let him merge a national bank with a trading monopoly in Louisiana and use the shares to absorb the crown's debt. Shares went from five hundred livres to around ten thousand, the word millionaire dates from the episode, and the whole structure fell apart within a year when holders began converting paper to coin. Law fled the country and died poor in Venice. France did not attempt a national paper currency again for eighty years.

WHAT THE ERROR WAS MADE OF

A circular structure that looked like a mechanism. The bank printed notes, the notes bought the shares, the rising shares justified more notes, and each part was supported by the others with nothing underneath. Louisiana produced almost nothing in this period; the trade that was supposed to service the debt never existed. A system that generates its own demand is not a business, and the test is simple: if the price stops rising, does anything remain that earns money. Ask what earns money at a standstill.

083

"I can calculate the motions of the heavenly bodies, but not the madness of people."

attributed to Isaac Newton, 1720

WHAT HAPPENED

He bought back in. Newton was Master of the Mint and the most celebrated intellect in Europe. He held South Sea stock and sold at a good profit early in 1720. Then he watched it continue upwards, bought back in near the top, and lost something like twenty thousand pounds, an enormous sum. The remark is not recorded in his lifetime and first appears more than a century later, so it belongs among the attributed rather than the documented. The losses are real and are visible in his accounts.

WHAT THE ERROR WAS MADE OF

Selling correctly and then being unable to bear watching. He got the analysis right. The second decision was not an analysis at all. It was the particular pain of watching acquaintances grow rich on a thing you have just called overvalued. That pain has no intellectual defence, and being extremely clever is no protection against it, which is why this entry has survived for three centuries as the standard warning. The dangerous moment in any mania is not the beginning. It is the point after you have got out and it keeps going up. The danger comes after you sell.

084

Railway shares are a sound investment for the ordinary saver, and the lines now projected will all be built and will all pay.

Charles Mackay, 1845

WHAT HAPPENED

He had written the manual. Mackay had published *Extraordinary Popular Delusions and the Madness of Crowds* in 1841, the founding text on financial mania and still the best known. In 1845, as editor of the Glasgow Argus, he wrote leaders encouraging investment in railway shares at the height of the railway mania. Parliament authorised lines that could not conceivably be built; the market broke in late 1845 and shares lost about two-thirds of their value; thousands of small investors were ruined. The later editions of his book do not mention it.

WHAT THE ERROR WAS MADE OF

Knowing the pattern in the abstract and not recognising it in the room. Mackay is the control experiment, which is why the entry is here. A man who had written the manual, with the tulips and the South Sea in his own index, stood inside the next one and saw a sound investment. Pattern recognition after the fact is nearly free. Applying it to the present, to something you like, in a year when everyone you respect is making money, is the whole difficulty. Mackay could not pay that price either.

085

Florida land cannot fall. They are not making any more coastline, and the whole of the north is coming down.

The Florida land boom, 1925

WHAT HAPPENED

It turned in the spring of 1926. Miami lots changed hands several times in a day, often on a ten per cent deposit and a signed binder, by young men who never intended to complete. Prices in some districts rose tenfold in three years. The market turned in the spring of 1926, the railways embargoed non-essential freight, and a hurricane in September finished it. Land in some subdivisions became worth less than the unpaid taxes on it.

WHAT THE ERROR WAS MADE OF

A true demographic story financed on terms that could not survive a pause. Florida's population really was growing and really did keep growing for the next century; a patient buyer in 1925 would eventually have done well. What killed people was the binder, which is to say the leverage. At ten per cent down, a fifteen per cent fall wipes you out, and the timing you have to be right about goes from decades to weeks. The asset was fine. The financing was the bubble, and that is true far more often than the stories admit. Look at the deposit, not the land.

086

Japanese land cannot fall in price. There is very little of it and the demand is permanent.

The Japanese land myth, 1989

WHAT HAPPENED

Land fell for fifteen years. At the peak, the land under the Imperial Palace in Tokyo was said to be worth more than the whole state of California. Japanese land was valued at roughly four times all the land in the United States. The Nikkei reached 38,915 on the last trading day of 1989. Land prices fell for the next fifteen years, some urban commercial land by more than eighty per cent, and the index did not see that level again for thirty-four years.

WHAT THE ERROR WAS MADE OF

A cultural belief mistaken for an economic law. The Japanese phrase for it translates as the land myth, and bankers, regulators and households held it alike. That is what made it dangerous. The belief was embedded in the collateral rules of the banking system, so land could not fall without taking the banks with it, and that is what happened. When everybody in a system believes the same thing about the value of what they lend against, the belief stops being an opinion and becomes structural. The correction is then not a price move but a solvency event.

087

"We believe the Dow should rise to 36,000 immediately, but to be conservative we expect it to take three to five years."

James Glassman and Kevin Hassett, 1999

WHAT HAPPENED

Twenty-two years. The book appeared in October 1999. The Dow peaked around 11,700 in January 2000, fell for three years, and first closed above 36,000 in November 2021, twenty-two years later. The argument was that investors had demanded too high a premium for holding shares and that the premium was about to disappear. It was a serious academic position and not a crank one.

WHAT THE ERROR WAS MADE OF

A theoretical adjustment given a date. The underlying claim, that the equity risk premium might be too high, was arguable then and is arguable now. What converted it into one of the most mocked titles in publishing was the decision to attach a level and a schedule. A valuation argument tells you about a long-run average and says nothing about which decade. The lesson is uncomfortable for anybody who forecasts. The more rigorous the reasoning behind a number, the more tempting it is to state when as well, and the when is almost never in the reasoning. Never attach a date to a valuation.

088

We can build a national brand for pet supplies before the money runs out.

Pets.com, 2000

WHAT HAPPENED

Nine months. The company floated in February 2000 and spent enormously on advertising, including a Super Bowl slot. It shipped heavy bags of dog food below cost, because the delivery economics did not work at the volumes it had. It liquidated nine months after listing. The sock puppet is remembered better than the company. Chewy, which sells the same goods online at scale, is worth billions, twenty years later, on the same idea and a workable cost base.

WHAT THE ERROR WAS MADE OF

Treating a unit economics problem as a scale problem. Every internal projection assumed that losses per order would turn positive with volume, which is true for some businesses and false for anything whose main cost is shipping weight. Nobody was fooled about the numbers; they were published. The belief was that growth would fix them, and growth made them worse in exact proportion. The question that separates this from

Amazon is not vision or nerve. It is whether the loss per unit falls as the units rise, and it is arithmetic anybody can check.

089

"Bitcoin will be worth a million dollars by the end of 2020."

John McAfee, 2017

WHAT HAPPENED

It ended 2020 at about $29,000. He first said five hundred thousand, then raised it, and attached a famously indelicate promise about what he would do if he were wrong. Bitcoin ended 2020 at about twenty-nine thousand dollars. In 2020 he said the prediction had been a device to attract new buyers into the market. That is either an admission or a further claim, depending on how much of anything he said you believe.

WHAT THE ERROR WAS MADE OF

A forecast made as a performance, in a market where the performance moves the price. That is what distinguishes speculative assets from every other subject in this book. The forecast is not an observation about the thing. It is an input to the thing, and a loud enough prediction can make itself briefly true. Nothing said publicly by a holder of a volatile asset about the future price of that asset is a forecast. It is a position being talked, and the appropriate response is to look at what they own. Look at what they own.

090

The old valuation measures do not apply to this class of asset, because the world has changed.

The received wisdom, 1637 to 2025

WHAT HAPPENED

It is said in some form near the top of all of them, and it has been true perhaps twice. Railways did change England. The internet did change everything. In both cases the technology delivered more than the enthusiasts promised and the shares still fell by around ninety per cent. A real transformation and an overpriced claim on that transformation are two different objects, and only one of them is what you own.

WHAT THE ERROR WAS MADE OF

A category confusion, embedded so deeply in the language that most people never see it. *Is this technology important* and *is this security a good purchase at this price* are unrelated questions with unrelated answers. Every mania in this chapter ran on the answer to the first being used to settle the second. If you take one habit from this book, take the habit of separating them, out loud, every time somebody explains that the rules have changed. They may well have. The price is still a number, and the number has to come from somewhere.

IV

PEOPLE

CHAPTER 10

Over By Christmas

Wars are begun by people who expect them to be short. This is close to a rule, and it is not a coincidence, because a war that everybody expected to be long and ruinous would usually be settled by negotiation instead.

The consequence is bleak and worth stating at the outset. The optimistic forecast is not an unfortunate accompaniment to the decision. It is a condition of the decision, and it is generated by the same people who want to go. The estimate comes from the part of the organisation with the strongest interest in the answer.

The specific error underneath most of these is a failure to model the other side as a thinking participant. Almost every entry here treats the enemy as a fixed object with known properties. A structure that will collapse when kicked. A population that will greet you. A front that cannot be crossed. What actually happens is that the other side observes what you do and changes what it does, and the second move of the war is never in the plan.

There is also a professional distortion that deserves naming. Military planning is done in the language of capability, which is a real and measurable thing: divisions, tonnage, range, days of fuel. Will is not measurable, and so it drops out of the arithmetic and reappears as an adjective. Nearly every mistake in this chapter is a correct calculation of capability with the other side's willingness to keep fighting set, silently, to zero.

Read this chapter with the previous one's warning attached. Hindsight makes these look like arrogance, and some of them are. Several are ordinary institutional forecasting failures with a body count. Every one of these estimates was produced by a professional staff with maps, timetables, fuel tables and an accurate count of the other side's divisions, and in almost every case the quantity that decided the outcome was the one quantity that appears nowhere in a staff table.

091

“You will be home before the leaves have fallen from the trees.”

Kaiser Wilhelm II, August 1914

WHAT HAPPENED

Four years and three months. The war killed something like seventeen million people. It was not only the Kaiser. The assumption of a short war was general across every staff in Europe. The German plan in particular required a decision in the west within six weeks, because the alternative was the two-front war they knew they would lose. Trench lines were established by that autumn and barely moved for three years.

WHAT THE ERROR WAS MADE OF

A plan whose success was assumed because its failure was unbearable. The six-week requirement was not a forecast; it was an input, and once the whole strategy depended on it, examining whether it was realistic became an attack on the strategy. This is the most dangerous structure in institutional life and it is not confined to armies. When a schedule is load-bearing, the schedule stops being evidence, and nobody inside can tell the difference. A load-bearing schedule is not evidence.

092

The Ardennes is not a dangerous sector. The terrain itself defends it.

Philippe Pétain, 1934

WHAT HAPPENED

The armour came through. France built the Maginot Line along the German border, an engineering achievement that worked exactly as designed. The Ardennes forest was left lightly held, on the grounds that armour could not pass through it. In May 1940 the German army put seven armoured divisions through the Ardennes, crossed the Meuse at Sedan, and reached the Channel in ten days. France capitulated in six weeks.

WHAT THE ERROR WAS MADE OF

A defence that told the attacker where to go. The line was not a failure of construction and the Germans did not breach it; they went round it, which is what a wall always invites. The deeper error is about dates. The impassability of the Ardennes was a judgement about the vehicles of 1934, never revisited as vehicles changed, and by 1940 it had the standing of a geographical fact. Any assumption of the form *this route is impossible* should carry the date it was last checked. Date every impossibility.

093

"One thing is certain: he missed the bus."

Neville Chamberlain, 4 April 1940

WHAT HAPPENED

He was wrong within the week. Chamberlain told a Conservative party meeting that Hitler had lost his opportunity by failing to attack in the first winter of the war. Germany invaded Denmark and Norway five days later, and France six weeks after that. The remark contributed directly to his fall, and he resigned on 10 May, the day the attack in the west began.

WHAT THE ERROR WAS MADE OF

Quiet read as weakness. Nothing had happened for seven months, and Chamberlain, who wanted the war to be winding down, took the absence of events as evidence that the moment had passed. Nothing visible was happening because preparation is invisible, and an inactive opponent is either beaten or getting ready, and the two look identical from outside. It is also a lesson about certainty as a verbal tic: the words *one thing is certain* are almost never attached to anything that is. Quiet is not the same as beaten.

094

The reports of an imminent German attack are British disinformation intended to draw us into war.

Joseph Stalin, June 1941

WHAT HAPPENED

One day. Warnings came from Richard Sorge in Tokyo, from Churchill, from his own border commanders reporting German reconnaissance, and from a German deserter on the night of 21 June. Stalin dismissed them all and forbade actions that might provoke. Germany attacked on 22 June with three million men, and the Soviet Union lost most of its air force on the ground and several million men in the first six months.

WHAT THE ERROR WAS MADE OF

A theory of motive that explained away every piece of evidence in advance. Stalin's reasoning was not stupid: Britain did want the Soviet Union in the war, and would have benefited from provoking one. Having adopted that frame, every warning became further proof of the plot, and the more sources agreed, the more coordinated the deception appeared. A belief that converts confirming evidence into supporting evidence for itself cannot be dislodged by information, and the only defence is to decide in advance what would count against it.

095

“We have only to kick in the door and the whole rotten structure will come crashing down.”

Adolf Hitler, 1941

WHAT HAPPENED

The door did not give. The Soviet Union absorbed the deepest invasion in modern history, moved its industry east of the Urals, and destroyed the German army over the following four years. Something like eighty per cent of German military casualties in the war were inflicted on the eastern front.

WHAT THE ERROR WAS MADE OF

A political prejudice used as a military estimate. The belief that the Soviet state was rotten and would collapse under pressure was ideological and held with total conviction. It formed no part of any analysis of Soviet manpower, industry or geography, all of which were available and pointed the other way. It also had the property of making planning unnecessary, which is why it was attractive. When a forecast removes the need to prepare for the difficult case, examine the forecast before you examine the case.

096

There is very little chance the Chinese will come in. The boys will be home by Christmas.

Douglas MacArthur, October 1950

WHAT HAPPENED

Six weeks. MacArthur told Truman at Wake Island in October that Chinese intervention was unlikely and would be slaughtered if attempted. Some three hundred thousand Chinese troops were already crossing the Yalu. They struck in late November, as he launched what he had called the home-by-Christmas offensive. They drove United Nations forces back down the peninsula in the longest retreat in American military history. The war ground on for another two and a half years.

WHAT THE ERROR WAS MADE OF

Reasoning about what he would do in their position. Chinese intervention looked irrational by MacArthur's measures, since their army was poorly equipped and the Americans held the air, and by that arithmetic he was correct. What he did not weigh was that a hostile army arriving at their border was intolerable at almost any cost. That is a fact about their interests, not their capability. Estimating an opponent by whether their move would be sensible for you is the most common analytical error there is. Their interests, not yours.

097

"We have reached an important point when the end begins to come into view."

William Westmoreland, November 1967

WHAT HAPPENED

Tet. Ten weeks later the Tet offensive struck more than a hundred towns and cities across South Vietnam simultaneously, including the grounds of the American embassy in Saigon. Militarily it was a defeat for the North, which lost enormous numbers and did not hold anything. Politically it destroyed the credibility of every optimistic American statement about the war, and Johnson announced within two months that he would not seek re-election.

WHAT THE ERROR WAS MADE OF

A metric that measured the wrong war. American assessment rested heavily on enemy body counts, which really were running in their favour, and which counted a war of attrition that the other side was not fighting. The North's objective was not to win battles but to outlast a democracy's patience, and against that objective a favourable exchange rate is close to irrelevant. Choose a measure that your opponent is not optimising for and you will keep winning by it until the day you lose.

098

A limited contingent, in and out within a few months, to stabilise a friendly government.

The Soviet Politburo, December 1979

WHAT HAPPENED

Nine years. Soviet forces entered Afghanistan on 25 December 1979 and left on 15 February 1989, nine years and seven weeks later. They lost around fifteen thousand dead, and any remaining belief at home that the leadership knew what it was doing. The war is generally counted among the causes of the Soviet collapse two years afterwards. The decision was taken by a very small group of elderly men, against the advice of the general staff.

WHAT THE ERROR WAS MADE OF

A small decision that was never re-examined as a large one. The initial commitment really was limited, and each increase was small next to what had already been spent. At no point was there a meeting at which anybody decided to fight a nine-year war. That is how nearly all long wars are entered. The defence against it is to fix in advance what result, by what date, would cause you to leave, and to write it down while leaving is still cheap. Write down the exit before you go.

099

"I really do believe that we will be greeted as liberators."

Dick Cheney, March 2003

WHAT HAPPENED

The easy part went to plan. The conventional campaign took three weeks and the regime fell. Then the army was disbanded, the ministries were looted, an insurgency began within months, and American forces remained for eight years, with a further intervention after that. Estimates of Iraqi deaths over the period run into the hundreds of thousands.

WHAT THE ERROR WAS MADE OF

Planning for the part that was understood. The invasion was modelled in enormous detail and executed almost exactly as designed, and the question of what happens on the third day was handled by a sentence about liberation. It was not an oversight in the ordinary sense: several agencies produced studies of the occupation, and those studies were unwelcome and were set aside. A plan that is superb about the phase you have practised and lyrical about the phase you have not is not half a plan. The lyrical part is where the outcome lives.

100

Kyiv in three days, the country in a fortnight.

The Russian General Staff, February 2022

WHAT HAPPENED

Five weeks. The plan was built around the assumption that resistance would be minimal and the government would fall or flee. Its execution showed the belief: columns on single roads without secure flanks, three days of rations, and parade uniforms carried for the ceremony in Kyiv. The advance on the capital was abandoned within five weeks. The war has continued for years, with casualties on a scale not seen in Europe since 1945.

WHAT THE ERROR WAS MADE OF

Intelligence that had been shaped by what the leadership wished to hear, feeding a plan with no capacity to be wrong. The estimate of Ukrainian will was not a mistaken measurement; it was an unmeasured quantity supplied by the people who wanted the operation. What makes it the modern version of every other entry here is the design of the plan itself. A force equipped for three days cannot discover that the assumption was wrong and adapt, because it has already spent everything it would need in order to adapt. Confidence is cheap in a forecast and enormously expensive in a logistics table. Three days of rations is a forecast.

CHAPTER 11

The Verdict Was In

Predicting what people will do collectively is the hardest forecasting there is, and it is the one everybody attempts, because the answers are entertaining and nobody keeps score.

Some of this chapter is about polling, which is worth defending before it is criticised. A well-run poll is one of the few genuine instruments in social life, and over the past century it has been right far more often than it has been wrong. When it fails it fails in ways that are understood afterwards, and there are three of them. The sample is not the population. The respondents do not tell the truth. And the people who decline to answer are not a random subset of everyone. Every failure below is one of those three.

The larger part of the chapter is about something else, which is the confusion between a measurement and a forecast. A poll measures a stated intention now. Turning it into a prediction of an event later requires a model of who will actually go, what the undecided will do, and how the numbers translate into seats. Almost all the famous errors live in that second layer, and not in the polling. The instrument was reporting honestly. The interpretation was doing the work.

Then there is the probability problem, which is new and getting worse. A forecast of seventy per cent is not a prediction, it is a distribution, and it is compatible with the other thing happening three times in ten. Published in a headline it reads as a promise. When the three-in-ten happens the audience concludes the forecast was wrong, and that is not the audience's mistake. It belongs to whoever chose to publish a probability to people who were never offered a way to be right about it. The instructive thing about the entries here is how many were produced by the best available method, applied competently, by people who published their workings, and were then undone by a single assumption so ordinary that nobody in the room had thought to write it down.

101

War between the industrial powers has become futile, and therefore will not happen.

Norman Angell, 1910

WHAT HAPPENED

Four years. Angell's book *The Great Illusion* sold in enormous numbers and was translated everywhere, and the popular version of its argument is the sentence above. What he wrote was that conquest could no longer be economically profitable in an interdependent world, and that this made war irrational. It was true in 1914 and it is true now. He did not say it was impossible, and he spent the rest of his life saying so. He won the Nobel Peace Prize in 1933.

WHAT THE ERROR WAS MADE OF

Nothing, by him, and a great deal by his readers, which is why the entry is here. A careful conditional claim, that war would be unprofitable, was compressed by an audience that wanted reassurance into an unconditional one, that war would not occur. This happens to almost every popular book with a thesis, and it is worth knowing from both ends. As a reader, check whether the striking claim is the author's or the summary's. As a writer, assume the qualification will be removed.

102

Landon will take fifty-seven per cent of the popular vote.

The Literary Digest, 1936

WHAT HAPPENED

He lost forty-six states. Franklin Roosevelt took just under sixty-one per cent and every state but two. The Digest had polled on a scale nobody has attempted since, mailing ten million ballots and receiving nearly two and a half million back. It had called the previous five elections correctly. George Gallup, with a sample of about fifty thousand, called it right, and also predicted in advance what the Digest's error would be. The magazine folded within two years.

WHAT THE ERROR WAS MADE OF

A sample drawn from lists of people who had telephones, cars and magazine subscriptions, in the middle of the Depression, when those things divided the country almost exactly along the lines of the vote. It is the founding lesson of survey research: size does not fix bias. Two and a half million responses from the wrong population are worse than fifty thousand from the right one, because they come with a confidence that the small honest sample does not.

103

"I believe it is peace for our time."

Neville Chamberlain, 30 September 1938

WHAT HAPPENED

Eleven months. Chamberlain returned from Munich having agreed to the German annexation of the Sudetenland, and read the declaration to a crowd in Downing Street. Germany occupied the rest of Czechoslovakia in March 1939 and invaded Poland in September. It should be said that the year bought was used for rearmament, and that the argument about whether Britain was better placed in 1939 than in 1938 is still live among historians.

WHAT THE ERROR WAS MADE OF

A concession read as a settlement because it had been paid for. Chamberlain had given something real and received a signature, and the natural human accounting is that a costly agreement is a solid one. The other party's incentives had not changed at all, and a promise is worth what it is worth to the promiser to keep it, not what it cost you to obtain. Any deal whose durability rests on the other side's satisfaction rather than on their interest is a truce with a date on it.

104

DEWEY DEFEATS TRUMAN

The Chicago Tribune, 3 November 1948

WHAT HAPPENED

Truman won. The Tribune, working against a printers' strike and an early deadline, went to press on the strength of polling that had shown Dewey comfortably ahead all year. Gallup had stopped polling in mid-October on the reasoning that voters do not change their minds in the last fortnight. They did. Truman was photographed the next morning holding the paper up, delighted, and it is the most famous newspaper front page ever printed.

WHAT THE ERROR WAS MADE OF

An assumption inside the method rather than an error in the data. Everybody's polling was accurate about the moment it was taken; the fatal belief was that opinion was settled, so measurement could stop. That assumption was inherited from a period when it had been true and was never tested again. In any instrument, look for the thing that is not measured because somebody once decided it does not vary.

105

The Soviet Union faces serious economic difficulties but remains stable, and will be a superpower rival for the foreseeable future.

Western intelligence assessments, 1988

WHAT HAPPENED

Three years. The Berlin Wall opened in November 1989, the Warsaw Pact dissolved in 1991, and the Soviet Union ceased to exist on 26 December 1991, three years later. Western agencies had documented the economic decline in detail and some analysts had called the system unsustainable. What essentially nobody produced was a forecast that the state itself would dissolve, peacefully, within five years.

WHAT THE ERROR WAS MADE OF

An institution built to answer a different question. The apparatus had been built over forty years to count divisions, warheads and factory output. It did all of that well. None of it measures whether the people running a country still believe in it. Legitimacy is not observable by the instruments an intelligence service owns, so it was not in the estimate. Its absence was invisible, because every quantity that was in the estimate looked normal until the week it did not.

106

"The Wall will still be standing in fifty and even in a hundred years."

Erich Honecker, January 1989

WHAT HAPPENED

Ten months. Honecker was removed by his own party in October and the crossings were opened on 9 November after a press conference in which a spokesman misread a note about travel regulations. The state he led ceased to exist within a year.

WHAT THE ERROR WAS MADE OF

Confusing an absence of visible dissent with the presence of consent. East Germany in January 1989 was quiet. It was quiet because dissent was punished, which tells you nothing about what people believe and everything about what they will say. Any system that suppresses the expression of a preference destroys its own ability to measure it. The leadership finds out last, being the group most insulated from the one thing that would inform it.

107

Liberal democracy is the end point of ideological evolution and the final form of human government.

Francis Fukuyama, 1989

WHAT HAPPENED

History resumed. The next three decades produced an authoritarian and economically successful China, religious politics on a scale nobody had forecast, and a global financial crisis. Then a democratic recession measured across most indices, and major wars of conquest in Europe. Fukuyama's essay is far more careful than the slogan. Its original title ends in a question mark, and it worries out loud that the end of history might be boring enough to restart it. He has revised the argument at length since and has been unusually willing to say where it was wrong.

WHAT THE ERROR WAS MADE OF

A trend at its maximum mistaken for a destination. The essay was written in the most favourable year for its thesis in modern history. That is not a coincidence. It is when such a thesis becomes thinkable. The general form is the one from chapter four, in political dress. A period of consolidation looks like completion from the inside, and the anomalies that will undo it are already present, filed as exceptions.

108

The election is too close to call, and a hung parliament is the likeliest outcome.

The British polling industry, 1992

WHAT HAPPENED

They were out by eight and a half points. The Conservatives won by nearly eight points and took more votes than any party in British history. The final polls had understated them by about eight and a half points, the largest polling error in a British general election. The Market Research Society's inquiry found two causes: samples that were unrepresentative in a way the quotas did not catch, and respondents who intended to vote Conservative and would not say so. The industry rebuilt its methods and then made a related error in 2015.

WHAT THE ERROR WAS MADE OF

A measurement that changes when the respondent knows what the answer signals. A poll assumes that reporting your intention is costless. Where a vote has become socially disapproved of in somebody's own circle, it is not, and the misreporting is systematic instead of random. It is the hardest problem in the field and it has no full solution. What it does have is a warning sign: whenever the answer to a question carries a social cost, treat the resulting numbers as a floor for the unpopular option.

109

"Romney will win by more than five points and take three hundred and twenty-five electoral votes."

Dick Morris, 2012

WHAT HAPPENED

He lost the argument and his slot. Barack Obama won by about four points and took three hundred and thirty-two electoral votes. Morris was a professional political consultant and television commentator. He made the prediction repeatedly and with escalating confidence, and dismissed the aggregated public polling, all of which pointed the other way, as skewed. He lost his television contract.

WHAT THE ERROR WAS MADE OF

A conclusion arrived at first and a methodology adopted to support it. The move used at the time was to reweight published polls by a turnout model of one's own choosing. It is a legitimate technique that will produce any answer you like, because the turnout assumption is exactly the unknown. When somebody rejects a body of data by adjusting a parameter that cannot be observed, the adjustment is the argument. Ask what the parameter would have to be for their answer to hold.

110

The probability of a Clinton victory is over ninety-nine per cent.

Sam Wang, 2016

WHAT HAPPENED

She lost the electoral college. Wang was a Princeton neuroscientist running one of the most respected election models. He had promised on air to eat a bug if Trump exceeded two hundred and forty electoral votes, and he ate a cricket on television afterwards, which is more than most people in this book did. Other models were less extreme: the most cautious of the well-known forecasters gave Trump around a three in ten chance, and was widely criticised at the time for being too generous.

WHAT THE ERROR WAS MADE OF

Treating state errors as independent when they are strongly correlated. Wisconsin, Michigan and Pennsylvania can be wrong for the same reason at the same time, and they were. Aggregating many state polls then does not shrink the uncertainty anything like as fast as the arithmetic suggests. It is the same failure as the mortgage ratings in chapter seven, in a different costume. It will keep recurring. Independence is the assumption that makes the mathematics tractable, and it is almost never true of human beings in the same country in the same month.

CHAPTER 12

When You Are the One Who Is Certain

Everything so far has been about other people. This chapter is about you, and it is the reason the book exists, because a collection of famous errors read for entertainment leaves you worse off than when you started. It makes you feel clever about hindsight and changes nothing you will do on Monday.

The estimates in this chapter are mostly ordinary. Not one was made by a fool. None required a scientific revolution to overturn. Every one was produced by professionals with an incentive to be right, using methods that had worked before. They are here because they are the version of the error you are actually going to make. You will not predict the end of physics. You will say six weeks.

Four things help, and they are all uncomfortable, which is why they are rarely used.

The first is the outside view. Ignore the details of your project and ask what happened to the last twenty things like it. This feels wrong, because you know things about your case that were not true of theirs, and that is exactly the feeling the method is designed to overrule. Everyone on all twenty of those projects also had reasons why theirs was different. The base rate beats your reasoning about your own case, consistently, and by a wide margin.

The second is the premortem. Before committing, gather the people involved and tell them it is two years from now and the thing has failed badly. Ask them to write down why. This gets you information that no request for concerns ever will, because it removes the social cost. Nobody is objecting. Everybody is explaining, and the doubts that would otherwise stay in the corridor arrive in writing.

The third is a date and a number. A belief that cannot be wrong on a particular day by a particular measure is not a belief you can learn from. *Transitory* and *soon* and *the end begins to come into view* are the sound of a claim arranging not to be scored.

The fourth is the cheapest and the one people skip. Find out who has already told you this will not work, and read what they said rather than remembering how they said it. Almost every disaster in this book had that person, and in most cases they are on the record, before the event, in the file. Take the four together and they amount to one instruction, which is to arrange in advance for the world to be able to tell you that you are wrong, at a moment when you can still act on it, through somebody whose job is not made harder by saying so.

111

Complete in 1963, for about seven million pounds.

The New South Wales government, 1957

WHAT HAPPENED

Ten years late. The Sydney Opera House opened in 1973 and cost about a hundred and two million Australian pounds, some fifteen times the estimate. The architect, Jørn Utzon, resigned in 1966 and never returned to Australia or saw the finished building. It is now among the most recognised structures on earth and is generally reckoned to have been worth it, which is the honest complication in every entry in this chapter.

WHAT THE ERROR WAS MADE OF

Construction begun before the design was solved. Work started in 1959, three years before anybody knew how to build the shells, in order to make the project irreversible before the opposition could gather. It worked, and it is why the building exists. The cost is that every subsequent decision was made under the constraint of foundations already poured for a different building, and the schedule was fiction from the first day. Starting early to prevent cancellation is a real strategy with a known price, and it should at least be chosen deliberately rather than described as a plan.

112

Development will cost around seventy million pounds, shared between the two countries.

The British and French governments, 1962

WHAT HAPPENED

Concorde cost somewhere over a billion pounds to develop, more than fifteen times the estimate in nominal terms. The treaty had been written without a break clause, so that neither government could withdraw. Around a hundred and fifty orders were expected from airlines around the world; fourteen aircraft entered commercial service, all with the two national carriers. It was a magnificent aeroplane and it never came close to recovering its costs.

WHAT THE ERROR WAS MADE OF

An estimate made when the political decision had already been taken. The figure was not the output of an engineering study; it was the number that would get the agreement signed, and everyone involved understood that raising it would kill the project. This is the standard mechanism behind large public overruns and it requires dishonesty from nobody. An optimistic estimate is selected because pessimistic ones do not get funded. So the projects that go ahead are systematically the ones that were most wrong about their costs.

113

"The Olympics can no more have a deficit than a man can have a baby."

Jean Drapeau, 1970

WHAT HAPPENED

Twelve times over. The 1976 Montreal games cost around one and a half billion dollars against an estimate of a hundred and twenty million. The stadium's tower was not finished until 1987. The debt was finally paid off in 2006, thirty years after the closing ceremony, by a tobacco tax. A cartoonist drew Drapeau pregnant and the drawing outlived the argument.

WHAT THE ERROR WAS MADE OF

A certainty expressed as a physical impossibility, which is worth flagging as a rhetorical form because it does specific damage. He did not say it was unlikely, or that he was confident. He chose an analogy from biology, and the effect of that construction is to make disagreement absurd instead of merely wrong. Language of that kind commits the speaker far past the evidence and closes the room, and the tell is that no number appears anywhere in the sentence.

114

Two point eight billion dollars, complete by 1998.

The Massachusetts Highway Department, 1985

WHAT HAPPENED

Five times over. Boston's Central Artery and Tunnel project, the Big Dig, was substantially finished in 2007 at about fourteen and a half billion dollars. With interest on the borrowing the total is usually put nearer twenty-four billion. A ceiling panel in one of the tunnels fell and killed a motorist in 2006. The road it replaced was an elevated expressway through the middle of the city, and by most accounts the result was worth having, at a price nobody would have voted for.

WHAT THE ERROR WAS MADE OF

Pricing the known work of a project whose difficulty was all in the unknown. Digging a tunnel is a well-understood activity with reliable unit costs. Digging one under a functioning city, beneath an elevated road that must stay open, through fill of unknown composition, next to utilities nobody has a map of, is a different activity. The estimate was built by costing the first one. When the novel part of a job is the interaction with everything around it, your unit costs are describing a job you are not doing.

115

Four point seven billion pounds, opening in 1993, carrying thirty million passengers a year.

Eurotunnel, 1987

WHAT HAPPENED

The Channel Tunnel opened in 1994 at around nine and a half billion pounds, roughly double the forecast. The traffic was the more serious error. The projections were missed by a wide margin for years. The ferry companies, which the model had treated as a fixed alternative, cut their prices and improved their service, and then low-cost airlines arrived. The original shareholders were nearly wiped out in a restructuring. The tunnel itself is an engineering triumph and now carries a great deal of traffic.

WHAT THE ERROR WAS MADE OF

A demand forecast that held the competition still. Every large infrastructure case study contains one. The model asked how many people would cross the Channel and how many would prefer a train, at the prices ferries then charged. A ferry operator facing an existential threat does not keep charging those prices. Any forecast that assumes your entry into a market will not change the behaviour of the people already in it is describing a world in which you did not build the thing.

116

The automated baggage system will open with the airport in October 1993.

The City of Denver, 1993

WHAT HAPPENED

The bags were shredded. The airport opened sixteen months late, at a reported cost of around a million dollars a day in delay, because the baggage system did not work: bags were shredded, misrouted and dumped. A manual system was built alongside it. The automated system was used in reduced form by one airline and abandoned entirely in 2005, after which the airport went back to conveyor belts and people.

WHAT THE ERROR WAS MADE OF

A subsystem on the critical path that had never been built at that scale, adopted late, with no fallback. Each of those alone is survivable. The combination is not, and the combination is common, because the ambitious component is the one added last and the one nobody wants to hedge against. The practical rule is simple and widely ignored: if a single unproven element can stop the whole thing opening, either prove it first or build the boring version in parallel.

117

Between ten and forty million pounds.

The Scotland Office, 1997

WHAT HAPPENED

The Scottish Parliament building at Holyrood was completed in 2004 at four hundred and fourteen million pounds, roughly ten times the top of the range and forty times the bottom. Lord Fraser's inquiry found that the figure had been for a different site, a different brief and a different building, and that it was never withdrawn as those things changed.

WHAT THE ERROR WAS MADE OF

A number that outlived every assumption it was built on. This is the most common version of the error in ordinary working life, and the least dramatic. Nobody lied and nobody was reckless. An early estimate for one thing became the public figure, and the project changed underneath it in a dozen legitimate ways. No moment ever arrived at which somebody was required to say that the old number no longer meant anything. Estimates need a stated basis, and when the basis changes the estimate is void, not adjustable.

118

Opening in October 2011.

Berlin Brandenburg Airport, 2006

WHAT HAPPENED

It stood empty and lit. It opened in October 2020, nine years late, at around three times the budget. The delays ran through fire safety systems that did not work, cable installations that did not meet code, and a period in which the terminal's lights could not be switched off. Berlin ran two ageing airports for a decade while a finished-looking building stood empty, heated and staffed.

WHAT THE ERROR WAS MADE OF

An opening date defended by declaring the problems solved rather than solving them. The pattern in the public record is a sequence of announcements, each confident, each followed by the discovery that the underlying issue was still there. Once a date has been announced enough times, the cost of moving it exceeds the cost of asserting it again, and the organisation begins managing the announcement instead of the building. Notice which one is being worked on. It is usually visible from outside.

119

Thirty-three billion dollars, San Francisco to Los Angeles by 2020.

The California High-Speed Rail Authority, 2008

WHAT HAPPENED

It is still not built. The estimate that voters approved in a referendum has risen several times over and no through service exists. The segment under construction runs through the Central Valley, between cities that were not the point of the project. Work continues. Whether it eventually gets built is still open, which is why it is the last of the projects here. It is the one you would have to judge without hindsight.

WHAT THE ERROR WAS MADE OF

A number produced for a ballot rather than for a builder. An estimate that has to win a vote is not the same document as one that has to guide procurement. Once a public has approved the first, revising towards the second looks like betrayal instead of arithmetic. If you want to know how good a forecast is, look at what it was for. A figure that had to persuade somebody is not a measurement, whatever it is dressed as.

120

Our considered forecasts about political and economic events are substantially better than chance.

The experts, 1984 to 2003

WHAT HAPPENED

They were not. Philip Tetlock spent twenty years collecting some twenty-eight thousand specific, dated, checkable predictions from nearly three hundred professionals whose job was to know: academics, officials, journalists and analysts. Scored against the outcomes, the average expert performed a little better than random and worse than simple statistical rules. Two findings are worth more than the headline. Those with the strongest organising theories did worst, and did worse the more famous they were. And the experts who did best held loose, provisional, patched-together views, updated often, and were the ones nobody wanted on television.

WHAT THE ERROR WAS MADE OF

Not knowledge, which they had, but the conversion of knowledge into confidence. Expertise makes you better at explaining and no better at forecasting, and it makes you far more certain, which is the combination that produced every entry in this book. Tetlock's later work is the encouraging half. Forecasting is a skill and it can be measured, and people trained to break questions down, start from base rates and update in small increments get much better within months. Almost nobody does that, because it involves saying seventy per cent out loud and being seen to move. One last thing. You have now read a hundred and twenty examples of people being certain, and the feeling you have about your own next estimate has probably not changed at all. That feeling is the subject of the book.

A Note on Quotation

A book about people being confidently wrong has an obligation. It is not to be confidently wrong itself. This note is about how the lines here were treated and which of the famous ones did not make it in. The short version is that a quotation which is short, clean, self-contained and ends on an ironic sting has usually been improved by somebody, and the improvement is nearly always made a few decades later by a person who was not there and had no reason to be careful.

The three forms used

A line **in quotation marks** is as recorded. The source is contemporary or close to it, and the wording is the one accepted by people who have looked.

A line **in bold, without quotation marks**, is a summary of a documented position rather than a quotation. It is used where somebody's view is beyond dispute but the exact words are not. It is also used where a position was set out over pages and not in a sentence. It is also used for institutional positions, which are almost never expressed in a quotable line by an individual.

A line marked **attributed to** is one where the sentiment is widely reported and the sourcing is thin. Doubt is on the record, not hidden. The entry says so and the note under it usually explains what is solid. Newton on the madness of people, Foch on aeroplanes and the Decca verdict on the Beatles are all in this category: the underlying events are documented and the famous wording is not.

The famous ones that are not here

Several of the best-known quotations in this genre appear to be inventions. A collection like this is where they breed. They are left out. Since their absence may look like an oversight, here is why.

”I think there is a world market for maybe five computers.” Attributed to Thomas Watson of IBM in 1943. No source has ever been found, in IBM’s own archives or anywhere else, and the earliest appearances are from the 1980s.

”Everything that can be invented has been invented.” Attributed to Charles Duell, commissioner of the United States patent office, in 1899. It was traced to a magazine humour column. Duell said the opposite in his annual report, which forecast that the century to come would astonish those who lived in it.

”There is nothing new to be discovered in physics. All that remains is more and more precise measurement.” Attributed to Lord Kelvin in 1900. There is no source. Kelvin’s real address of that period identifies two unexplained clouds over the theory. They turned out to be relativity and the quantum. The genuine remark is nearly the opposite of the fake one. Michelson said something close to the fake in 1894 and is quoted here instead, because he actually did.

”640K ought to be enough for anybody.” Attributed to Bill Gates. He has denied it repeatedly and no contemporary record exists.

”This telephone has too many shortcomings to be seriously considered as a means of communication.” Presented as a Western Union memo of 1876. The memo has never been produced. The decision to decline the patent is well documented and is in the book on that basis.

”Who the hell wants to hear actors talk?” Attributed to Harry Warner, whose company then made the first talking picture. The sourcing is poor and the sentiment is contradicted by what his company did.

Why the fakes cluster here

Two reasons, and both are useful to know about beyond this subject.

The first is that a fabricated quotation is better written than a real one. Real people hedge and qualify. They say things that need a paragraph of context. The fake is short, clean and self-contained, and it has an ironic sting that a genuine remark almost never has. If a quotation is perfect, be suspicious of it in proportion to how perfect it is.

The second is that famous people act as magnets. A line about computers drifts towards the most famous computer man, and a line about physics towards the most famous physicist. The drift happens a few decades after the fact, by which point nobody can trace it. This is a documented phenomenon in quotation research, and Mark Twain, Winston Churchill and Albert Einstein have absorbed most of the unclaimed epigrams of the English language between them.

Dates and revisions

Dates are when the thing was said or published. Not when it became famous. Several people in this book changed their minds, publicly and to their credit, and where they did the entry says so. A collection that quotes a man’s worst sentence and omits his retraction is doing the thing this book is supposed to be against.

Where the underlying facts came from

Ordinary ones. Contemporary newspapers and journals where they exist, official inquiries and accident reports for the engineering entries, published memoirs and the standard histories elsewhere. Nothing here required special access. Anyone could repeat it. The point of saying so is that the checking is available to anyone. If you find an error in this book, the machinery for correcting it is the machinery the book is about.

Index of People

UNSIGNED AND INSTITUTIONAL

❧

Confident and Wrong

Three hundred and fifteen entries across twenty-one chapters,
drawn from a hundred and fifty-one sources.

Set in EB Garamond, cut by Octavio Pardo
after the types of Claude Garamont and Robert Granjon.